Republic of the Philippines
North Eastern Mindanao State University
Rosario, Tandag City, Surigao del Sur 8300
Telefax No. 086-214-4221
Website: www.nemsu.edu.ph
College of Information Technology Education

IOT-BASED RAGDOLL AS BIRDS AND PEST DETERRENT

An Undergraduate Thesis

Presented to the Faculty of College of Information Technology Education

NORTH EASTERN MINDANAO STATE UNIVERSITY

Rosario, Tandag City, Surigao Del Sur

In Partial Fulfillment

Of the Requirements for the Degree

Bachelor of Science in Computer Science

By:

GODINEZ, JOMAR F.

GONJORAN, JEICER JOHN D.

LINAZA, LEA MAE V.

MACA, JEN-JEN M.

MACHECA, JEROME P.

MAGALLANES, RIALYN N.

JUNE 2023

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ACKNOWLEDGEMENT

This study is the result of a year's worth of hard work, helped along by researchers and people who contributed their time, expertise, and effort. In some way to this study's conclusion. They would like to express their sincere gratitude to the following individuals:

To the All-Powerful God, creator, redeemer, and sustained, for giving us courage, strength, unwavering love, unimaginable blessings, wisdom, knowledge, and the ability to finish this study.

The researchers are deeply thankful to their adviser, **Mr. Michael E. Estal**, **MSCS** for his unwavering support, patience, understanding, and crucial direction and assistance in the improvement and completion of the article.

The researchers would like to express their heartfelt gratitude to **Dr. Catherine R. Alimboyong** for her dedication and strong attention, particularly her overwhelming attitude to support her students. Her prompt and rigorous counsel Investigation and intellectual advice have aided them in accomplishing this assignment to a large extent.

To the families, their deepest gratitude. Your words of support during difficult times were much appreciated. Knowing that you were willing to manage their household tasks while we finished our work gave us great comfort and relief.

DEDICATION

This study is fully dedicated to our revered parents, who have always provided us with moral, spiritual, emotional, and financial support. Our dear parents you are our source of inspiration and strength when we were about to give up.

To our relatives, mentors, friends, classmates, and brothers and sisters who advised and encouraged us to finish this study.

Last but not least, we dedicate this book to the all-powerful God. We are grateful to you for your direction, strength, mental power, protection, skills, and healthy life. We offer all of these to you.

ABSTRACT

Rag dolls are a form of mannequin or decoy commonly utilized in crop fields across the country, especially in the southern part of the country, and are normally human-like in form. As the time for harvest draws near, rag dolls, cloths, and ropes are hung throughout the rice fields to scare away birds. Rice fields are home to a variety of undesirable and destructive animals, including birds. In most cases, farmers repel these bird attacks with conventional tools like scarecrows and plastic ropes.

This study compares several approaches to using PIR sensors to detect birds and selects the most accurate approach. It then applies this approach to automatically repel birds by using sound frequencies that birds dislike and that can repel pest birds. The design of the bird repellent device works by using computer vision techniques and PIR sensors to detect the thermal or bird infrared of each object in each frame. A micro controller then processes the data. The micro controller will use sound frequency to activate the actuator once the object has been detected by the PIR.

The goal of the research is to create prototypes based on the Internet of Things for monitoring and automatically controlling bird attacks to maximize crop yields in agriculture.

Keywords: *Agriculture, rice fields, rag doll, IoT-based, sensors*

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Chapter I

INTRODUCTION

Background of the Study

A ragdoll is like a mannequin or decoy, frequently human-shaped (humanoid) in the rice field and used all over the country, especially here in the Philippines. Rag dolls, clothes, and ropes are strung in maturing rice fields to frighten away birds as harvest time approaches. The owners of the rice field had erected a miniature dummy made of plant husks on a raised pole and strung various strips of fabric from ropes along a trail through the nearly ripe crops. In addition, farmers in rice fields used the rag doll as a prop while manually dragging the string and clanging a bell to protect their rice plants from dawn until dusk. Usually, the pest birds striked well before harvesting. Although the traditional ragdoll, which is usually seen standing in a field, is undoubtedly a graceful decoration that signified the beginning of autumn, from a more practical point of view, it has evolved into a thing of the past. Birds quickly grew accustomed to a stationary dummy and began to disregard it as unthreatening. As a result, modern ragdolls had taken the place of traditional ones.

In the study by Krol et al. (2019), it was said that the farmers on this farmland had faced several challenges due to the lack of new technologies, including income damage. Any replacement technique for manual effort that used

technology to bring out the tasks at hand is recognized as an automation system. This system was efficient in preventing animals from entering the agricultural fields and keeping them away. The study by Haque et al. (2021) stated that the high cost and complexity of IoT device scans also disappointed farmers with limited resources, especially those who lived in remote areas without adequate internet access or infrastructure, both of which were required for advanced monitoring systems to function. Moreover, Pradeepa et al. (2020) shared that auditory techniques are thought to be relatively effective despite being susceptible to habituation and therefore only of short-term benefit.

One of the difficulties stated by many farmers was that they always had a problem with hungry birds, especially when the rice is in the maturity phase. The main issues for coverage and protection are the environment in rice fields, bird attacks, and automatic pest management. One of the many animals that were a nuisance or destructive and could be found in rice paddies (Chan et al., 2019). The owner of the rice field had been making rag dolls to increase their yield because of the physical damage that hungry birds feeding on paddy fields caused. As an outcome, farmers are losing a lot of their paddy yield harvest in terms of quantity, quality, and the money they expected to make from their harvests. This was because the current traditional methods that these farmers use, like using a static rag doll, only worked for very small fields. Since birds consumed rice when they eat it, this was one of the reasons farmers fail.

Getting rid of insects, birds, and other pests on the farm is not an easy task for farmers. Farmers tend to spend most of their time deterring pests manually. Hence, it is important for the farmers to have a tool that allowed them to automatically deter birds and pests without human intervention through the IoT-Based Ragdoll as a Birds and Pests

Deterrent with the help of PIR sensors that could detect any object that had an infrared thermal. With the use of an IoT-based ragdoll, farmers would be notified of the presence of birds and pests around their rice fields through another feature of the innovative system, which is SMS notification. The owner or tenant of the rice field would receive a text message stating that the ragdoll is now active due to the birds or pests being detected, and the pests will be automatically deterred with the use of a sound that came from a rag doll.

Theoretical Framework

This research has both practical and theoretical impetus since the IoT is still in its infancy, yet it is considered by many to be the most important technology initiative of today. The study of White (2012) stated that using a naive theory of impetus, the object determined the type of motion it exhibits and gradually loses it over time.

Figure 1 shows the flow of the theory of impetus that were related to our study, in which an IoT-based ragdoll as birds and a pest's deterrent is an object

that played the motion of the pest birds. The theory of impetus stated that the objects that were caused to move by other objects acquired force that determines the kind of motion exhibited by the object, and that force gradually dissipates over time.

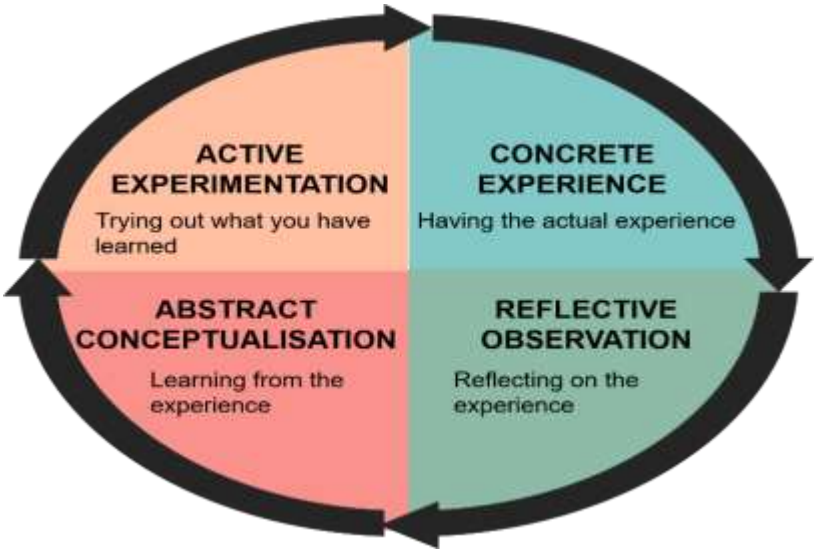

Figure 1. Naive Theory of Impetus

Active Experimentation - This stage of experiential learning was related to the concept of scientific experimentation, in which an individual formed a hypothesis based on existing ideas and tests the validity of these ideas in a structured experiment

Concrete Experience - being involved in a new experience. Reflective observation: watching others or developing observations about one's own experience.

Reflective Observation - watching others or developing observations about one's own experience. Abstract conceptualization: creating theories to explain observations. Active experimentation: using theories to solve problems, make decisions.

Abstract Conceptualization - In this stage, learning involves using theories, logic and ideas, rather than feelings, to understand problems or situations. Typically, the learner relies on systematic planning and develops theories and ideas to solve problems.

Prototype Model

This research obtained a prototype model that required the development of actual software; a working prototype of the system should be built, tested, and reworked until an acceptable prototype is achieved (Martin 2022). It also created a base to produce the final system or software. It worked best in scenarios where the project's requirements were not known in detail. It is an iterative, trial-and-error method that took place between developer and client.

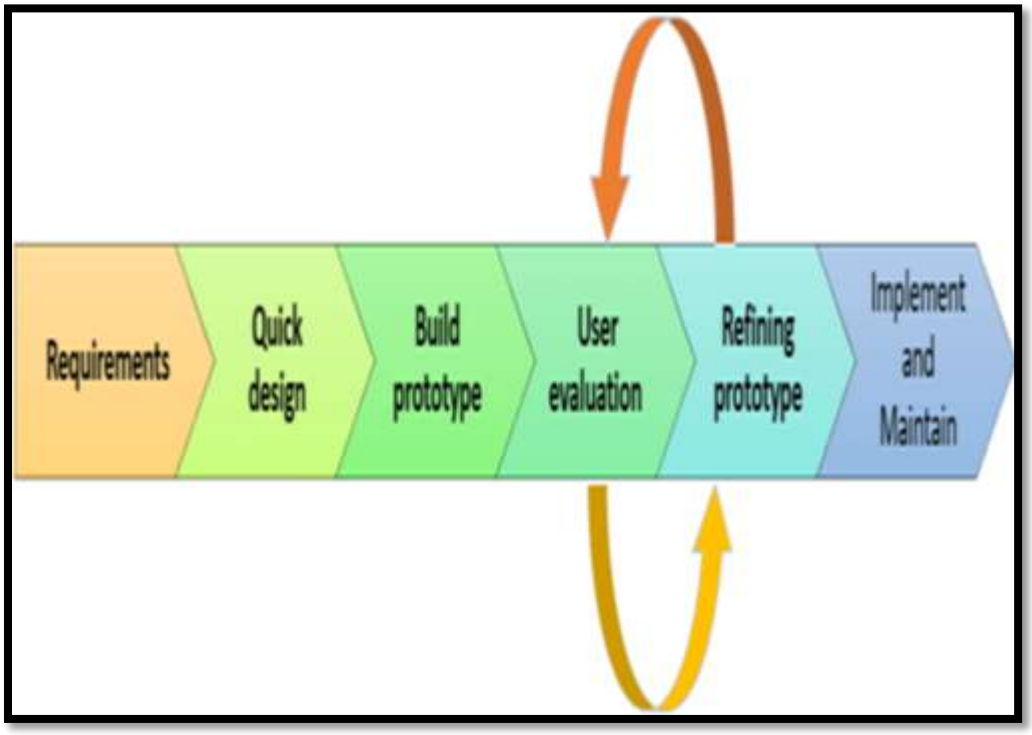

Figure 2. Prototype Model

Requirements Gathering and Analysis: A prototype model started with requirement analysis. In this phase, the requirements of the system were defined in detail. During the process, the users of the system were interviewed to know what their expectations are from the system.

Quick Design: The second phase is a preliminary design, or a quick design. In this stage, a simple design of the system was created. However, it was not a complete design. It gave a brief idea of the system to the user. The quick design helps in developing the prototype.

Build a Prototype: In this phase, an actual prototype was designed based on the information gathered from the quick design. It is a small working model of the required system.

Initial User Evaluation: In this stage, the proposed system was presented to the client for an initial evaluation. It helped to find out the strengths and weaknesses of the working model. Comments and suggestions were collected from the customer and provided to the developer.

Refining the Prototype: If the user is not happy with the current prototype, it needed to be refined according to the user's feedback and suggestions.

Implement, Product, and Maintain: Once the final system is developed based on the final prototype, it is thoroughly tested and deployed to production. The system undergoes routine maintenance to minimize downtime and prevent large-scale failures.

Conceptual Framework

The proponents used an IPO chart to present and describe the function of the proposed IoT-Based Ragdoll as Birds and Pest Deterrent system.

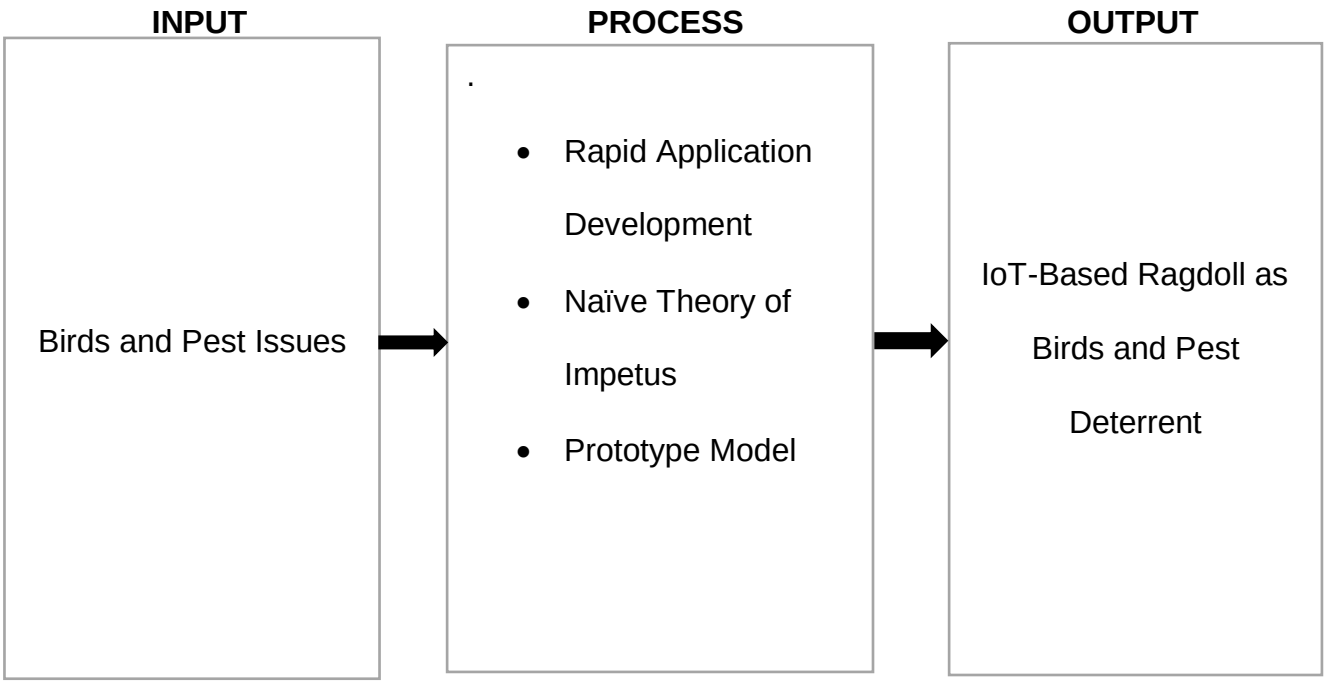

Figure 3. Conceptual Framework of the Study

Statement of the Problem

This statement gave an overview of the topic covered in the paper as well as determines the effect of the IoT-Based Ragdoll as a bird and pest deterrent on the owners of the rice field.

Specifically, the study aims to answer the following questions:

1. How could birds and other pests be prevented from entering farmland?
2. How could farmers benefit from utilizing the internet of things?
3. What is the level of satisfaction of the system in terms of the following dimensions?

   3.1 Functional Stability

   3.2 System Usability

   3.3 System Reliability

   3.4 Performance Efficiency

Scope and limitation of the Study

The scope of this research was limited for the following reasons: This IoT-based ragdoll was used as a bird and pest deterrent, focusing on objects that had an infrared sensor, and this automated ragdoll will be used during the harvest season of the rice field. Also, this system depends on the specific sites. The PIR sensor would only consider those objects that have an infrared. The range of the PIR sensor is around 5–7 meters in width and 5–7 meters in height in concave form. This system utilizes four PIR sensors in one Ragdoll rice field that has 1.5 hectares of rice fields. After the object is detected by the PIR, the micro controller would activate the actuator in the form of sound. Another feature of this study is that the owner would receive an SMS notification as a monitoring system, which would be implemented by loading the SIM card with a postpaid SIM card.

The limitation of this study is that those birds that had an infrared wavelength, when it comes to identifying the pest and birds that attacked the rice field, considered it safe when it is still in the tillering stage because they consumed the army worms. Army worms are caterpillars that attack rice. They're were at least three species of army worm that attack rice in Asia. These are the rice swarming caterpillar, the common cutworm, and the rice ear-cutting caterpillar. They can also cut the base of rice panicles off. The birds posed a threat due to their 5% contribution to crop production losses during the rice field's

maturity phase. In pest, we have our limitations; we could only focus on the insect that has infrared since our PIR Sensor only sensed the thermal of the subject. Natural enemies like damselflies, dragonflies, spiders, and mirid predators keep most of the insect pests of rice in Sagbayan at low levels. Indeed, their presence indicated the health of the rice field and lowers the likelihood of rice pest outbreaks. The rice black bug can't be detected by our automated ragdoll because these were usually detected when they gather around street lights and interact with humans in their houses. They emit a nasty smell when disturbed. In the field, they tend to congregate and feed by sucking the sap of the rice plants.

Significance of the Study

The study's findings indicated that auditory approaches were likely to benefit a variety of people and provide them with information that is relevant to them.

Farmers. This study would help the farmers to use this system in their rice fields, so they are able to do their tasks in farming easily.

Community. This study would inform the community that modern science and technology are prevalent today.

Department of Agriculture. This study will help the Department of Agriculture get some ideas about smart farming that will help farmers reduce their hardships in farming.

The Future Researchers. This study would serve as a guide for further studies to be conducted in the future. The study could also be opened during the development of this study. They would be able to use these data to get ideas and references if they were planning to conduct the same study.

Definition of Terms

The proponents of this study selected significant key terms to facilitate comprehension of the issue.

Arduino Mega - is a microcontroller board based on the ATmega2560. It has 54 digital input/output pins (of which 15 can be used as PWM outputs), 16 analog inputs, 4 UARTs (hardware serial ports), a 16 MHz crystal oscillator, a USB connection, a power jack, an ICSP header, and a reset button.

Arduino Nano - is a small, complete, and breadboard-friendly board based on the ATmega328 (Arduino Nano 3. x). It has more or less the same functionality of the Arduino Duemilanove, but in a different package. It lacks only a DC power jack, and works with a Mini-B USB cable instead of a standard one.

Buzzer - An Arduino Buzzer is basically a beeper. The Arduino buzzer is a device that produces sound when an electric current is passed through it. The Arduino buzzer can be directly connected to the Arduino and produce different tones by giving different frequency electric pulses to the buzzer.

Infrared Waves - enables your Arduino to receive and decode signals from a TV remote control. It is common to find TV remote controls that use infrared (IR) light to send encoded messages to the television. IR light wavelength is normally between 930 and 950 nm, which makes it invisible to humans.

IoT-Based - describes the network of physical objects “things” that are embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet.

nrf24L01 - is a single-chip radio transceiver for the worldwide 2.4 - 2.5 GHz ISM band. The transceiver consists of a fully integrated frequency synthesizer, a power amplifier, a crystal oscillator, a demodulator, a modulator, and an Enhanced ShockBurst™ protocol engine.

PIR Motion Sensor - allows you to sense motion, almost always used to detect whether a human has moved in or out of the sensor's range

.

Rag Doll - scarecrow that auto-detects bird sound and operates its arms using a motor along with shouting to scare away birds and animals for good.

SD Card Reader Module - allows you to communicate with the memory card and write or read the information on them. The module interfaces in the SPI protocol.

Sensors - Broadly speaking, sensors are devices that detect and respond to changes in an environment. Inputs can come from a variety of sources such as light, temperature, motion, and pressure. Sensors output valuable information and if they are connected to a network, they can share data with other connected devices and management systems.

SIM 800L - is a miniature cellular module which allows for GPRS transmission, sending and receiving SMS and making and receiving voice calls.

Smart Farming - Smart farming is an emerging concept that refers to managing farms using technologies like IoT, robotics, drones, and AI to increase the quantity and quality of products while optimizing the human labor required by production

Chapter II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter introduces the review of related literature and studies which provide relevant knowledge and information for enriching and giving light to the present study. This chapter has two sections such as related literature and related studies where each section is divided into two subsections such as local and foreign to facilitate the comprehensive gathering of data.

Related Literature

Foreign

Nowadays, the natural enemies of crops are birds, insects, and pests, which lower the annual income of our farmers. Recent developments in collection, computation, and communication had expanded the way data was reported in the intelligent Internet of Things (IoT). However, the diversity and complexity of data sources also imposed new trust challenges in the data collection process since untrustworthy reporters tend to report false or even malicious data, which highlights the need to develop a novel methodology to solve such challenges (Gaur et al., 2021). With the help of IoT, it possible to deter pests using sensors instead of pesticide spray. Internet-of-Things (IoT) applications were being widely used in various fields to empower intelligent smart

cities. Due to the smart decisions made by IoT applications depending on the reliability of data collection,

it is pivotal to collect data from trust-sensing devices. However, identifying the credibility of sensor nodes to ensure the credibility of data collection is a challenging issue (Shen et al., 2021).

In addition, Rehman et al. (2021) stated that the single sensor systems had been unable to distinguish between the true and false presence of fire, such as smoke from a cigarette, which might cause the fire alarm to go off. Consuming energy all day long and being dependent on one sensor that might end with false alerts is not efficient and environmentally friendly. Hence, Terry et al. (2021) developed a wildlife monitoring system that utilizes passive infrared (PIR) sensors to detect the presence of wildlife within the sensor's field of view. The system was tested and evaluated in a wildlife reserve and was able to detect the activity of various wildlife species, including birds, rodents, and larger mammals.

Also, with our society apparently becoming more technology-obsessed day by day, business organizations would have to involuntarily adapt these innovations to be more efficient and productive. These technologies can further be used to detect any potential bad behaviors (also called counterproductive work behaviors) in organizations. Researchers were increasingly focusing on these behaviors due to their effect on the well-being of the members of the organization and the organization itself. The Internet of Things, abbreviated as

IoT, is a technology framework that brings technologies together to execute specific actions. It is one

of the most intriguing concepts that is attracting the attention of many organizations (Savitha et al. 2020).

Moreover, the use of scarecrows as a deterrent to birds and pests was effective, according to Krol et al. (2019). However, with civilization progressing and being characterized by an emphasis on economic efficiency and agricultural engineering in the present day, scarecrows are disappearing from the rural landscape. Advanced electronic devices replace them, while scarecrows end up in museums and open-air museums called skansens, as well as being on display at local village festivals.

Local

Many types of local research were being published to show that the use of IoT can reduce people's effort and time. According to Valiente et al. (2018), that is an automated aquaponics system using Nile tilapia and Romaine lettuce with access to and control of pH level and temperature through the Internet of Things (IoT). Intel Edison was used as the microprocessor, which continuously send information on the aquaponics' status and adjusted them if the parameters fell below their optimal levels. The system could be monitored with the use of an Internet Protocol (IP) camera. A weekly comparison of the growth of plants and fish in automated aquaponics and traditional fishkeeping and hydroponics was

done, and growth in automated aquaponics was significantly greater than its traditional counterpart.

Ayaz et al. (2019) stated that the rapid emergence of Internet-of-Things (IoT)-based technologies redesigned almost every industry, including "smart agriculture," which moved the industry from statistical to quantitative approaches. Such revolutionary changes were shaking up the existing agricultural methods and creating new opportunities along with a range of challenges. This article highlighted the potential of wireless sensors and IoT in agriculture, as well as the challenges expected to be faced when integrating this technology with traditional farming practices. IoT devices and communication techniques associated with wireless sensors encountered in agriculture applications were analyzed in detail. What sensors were available for specific agriculture applications, like soil preparation, crop status, irrigation, and insect and pest detection, are listed. How this technology helped the growers throughout the crop stages, from sowing until harvesting, packing, and transportation, is explained. Furthermore, the use of unmanned aerial vehicles for crop surveillance and other favorable applications, such as optimizing crop yield, was considered in this article. State-of-the-art IoT-based architectures and platforms used in agriculture are also highlighted wherever suitable.

In addition, Kelly et al. (2020) added that the growth of IoT technology has driven interest in a wide range of health practices to improve population health

more specifically. (Sammeta, 2022) had reviewed that using the PIR sensor detect movement in general but do not reveal who or what moved. For that

reason, an imaging IR sensor is required. The term "passive infrared detector," or "PIR," is sometimes used to refer to PIR sensors. The term aloof alludes to the way that PIR gadgets don't emit energy for discovery purposes. They only function by detecting objects reflected or emitted infrared (radiant heat).

Moreover, with the exponential growth of the human race, which mean the growth of population, conventional or ancient farming methods were becoming unable to cope with the growth with satisfaction. Hence, advanced farming methods are much needed to meet the food needs of this growing number of people. In the recent few years, smart farming systems based on embedded systems and the Internet of Things (IoT) have gained popularity among people to enhance and complete food production (Mahbub 2020).

Related Studies

Foreign

Many studies were conducted with regards to the usage of the Internet of Things (IoT) in society. The Internet of Things (IoT) refers to networks with billions of physical devices for collecting, sharing, and utilizing data in the virtual world. Most IoT applications centralize security assurance when creating, authenticating, transferring, or deleting system components. However, the

centralization exposed its limitations in meeting the security needs of a rapidly growing number of things

worldwide. How to scale up the applications with assured security becomes a critical challenge (Viriyasitavat et al. 2019).

Insect pests inflicted damage to crop plants and deteriorated the quality of food grains and food products (Singh et al. 2018). With the development of information technology, the Internet of Things (IoT) and low-altitude remote-sensing technology represented by unmanned aerial vehicles (UAVs) were widely used in environmental monitoring fields. In agricultural modernization, IoT and UAV could monitor the incidence of crop diseases and pests from the ground micro and air macro perspectives, respectively (Gao et al. 2020).

Moreover, the Internet of Things is a platform where the design of the bird repellent device worked by utilizing computer vision techniques through PIR sensors. The devices became smarter every day; processing became intelligent every day; and communication became informative every day. While the Internet of Things was still seeking its own shape, its effects had already started to make incredible strides as a universal solution medium for the connected scenario (Ray 2018).

Local

Liu et al. (2018) found that mobile crowd sensing served as a critical building block for emerging Internet of Things (IoT) applications. However, the sensing devices continuously generated a large amount of data, which consumed a lot of resources (e.g., bandwidth, energy, and storage) and might sacrifice the quality of service (QoS) of applications. Prior work had demonstrated that there was significant redundancy in the content of the sensed data. By judiciously reducing redundant data, data size and load can be significantly reduced, thereby reducing resource costs, facilitating the timely delivery of unique, probably critical information, and enhancing QoS.

In addition, Emerita (2021) conducted a study that found that, as a concept, the Internet of Things (IoT) is exciting and fascinating. One of the challenging features of the IoT, however, is possessing a secure ecosystem that covered all the building blocks of the IoT design. Blockchain referred to the database holding a seamless and growing datasets. Generally, block chain had a distributive nature, indicating that there was no single or master PC controlling the entire chain. Instead, the involved nodes featured a copy of the entire chain. Understanding the diverse building blocks of block chains and IoT could help

determine the vulnerability areas in each unit. Accordingly, exploring the different technologies

required to address each of the emerging weaknesses is crucial to tackling the convergence issues between IoT and block chain technologies.

Moreover, Omitola et al. (2018) conducted a study in which they concluded that the flow of today's IoT supply chain is enormous. To produce an item in a global network of suppliers required a delicate dance of people, assets, systems, information, companies, and even governments. The opportunities for security vulnerabilities abound. In this paper, described the security challenges facing an IoT supply chain. The researchers first started by describing what an IoT endpoint is, its various types, and their concomitant security challenges. An IoT endpoint is an output of an IoT supply chain, and an enumeration of its security challenges is then used to describe what IoT attack surfaces are, together with the threats and vulnerabilities facing a typical IoT supply chain.

Synthesis

With the existence of the Internet of Things (IoT), most manual processes had been converted to digital technology. The rapid emergence of Internet-of-Things (IoT)-based technologies redesigned almost every industry, including "smart agriculture," which moved the industry from statistical to quantitative approaches. Such revolutionary changes were shaking up the existing

agricultural methods and creating new opportunities along with a range of challenges. It highlights the potential of wireless sensors and IoT in agriculture, as well as the

challenges expected to be faced when integrating this technology with traditional farming practices (Ayaz et al., 2019).

In addition, with our society apparently becoming more technology-obsessed day by day, business organizations would have to involuntarily adapt these innovations to be more efficient and productive. These technologies could further be used to detect any potential bad behaviors (also called counterproductive work behaviors) in organizations. Researchers were increasingly focusing on these behaviors due to their effect on the well-being of the members of the organization and of the organization itself. The Internet of Things, abbreviated as IoT, is a technology framework that brings technologies together to execute specific actions. It is one of the most intriguing concepts that is attracting the attention of many organizations (Savitha et al. 2020).

Through the advancement of technology, farmers and agriculturists want to innovate their bird, pest, and insect deterrent on the farm with the use of a sound that will irritate them by using an automatic sensor to detect those insects that cause the low harvest. The use of ragdolls as a tool that has embedded digital technology to repel insects would lessen the work of our farmers, and they can spend more of their time with their families.

Chapter III

METHODOLOGY

This section of the study presents the research methods, research design, research locale, research instrument/s, data gathering procedure, algorithm design, materials used in the study, as well as the statistical tools used in the treatment and analysis of data.

Quantitative Research Design

The design that was used in this study was the descriptive research design. Descriptive research design is a quantitative research method that attempts to collect quantifiable information to be used for statistical analysis of the population sample. The researcher used a survey collection method to gather the data. Survey data collection is any of a number of ways in which data can be collected for a statistical survey.

The Rapid Application Development (RAD) methodology is a flexible way to quickly create and deploy hardware applications. The RAD method is effectively designed to adapt to changes and new inputs like features and functions, updates, etc. It is designed to be flexible to changes and to accept new inputs, like features and functions, at every step of the development process.

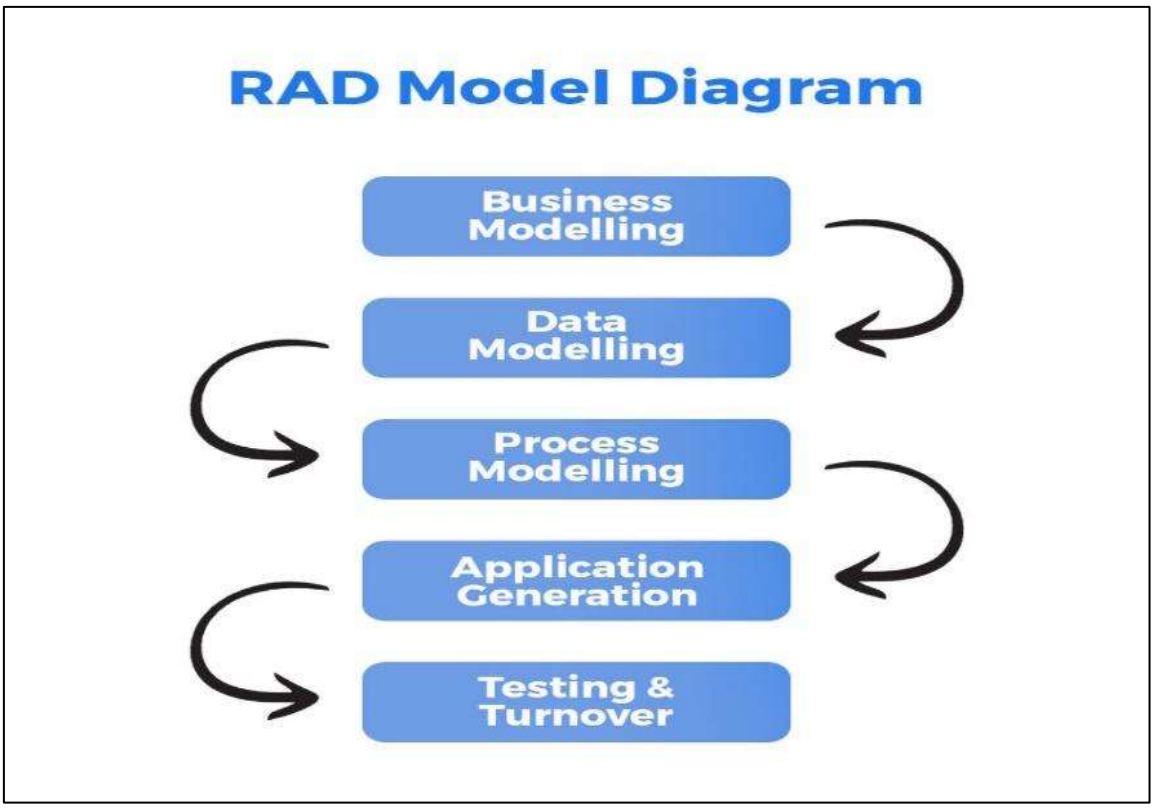

Figure 4. Rapid Application Development

Business Modeling – First, the researchers would find out how information flows throughout the study. This started with a broad analysis of the study by the researchers to uncover how vital information and data were obtained, how they're processed, where they're distributed, when all that happens, and what can slow it down.

Data Modeling – After the researchers pulled the information together, it was time to start analyzing it. To do that, the researchers identified and refined key data sets across the study, then group them in ways that might be useful later.

Process Modeling – It's time to take what had been learned and see where it all fits. That means beginning to understand the flow of information around key objectives and how that data supports specific system functions.

Application generation – This is when the concept started to become a reality and the researcher-developer team began to actually construct the system.

Testing and Turnover – Now it’s time to see what the users think. Here, prototypes were tested separately, so users can scrutinize each new component and examine it carefully to identify issues.

System Flowchart

This is how the IoT-Based Ragdoll as Birds and Pest Deterrent works base on the diagram shown below:

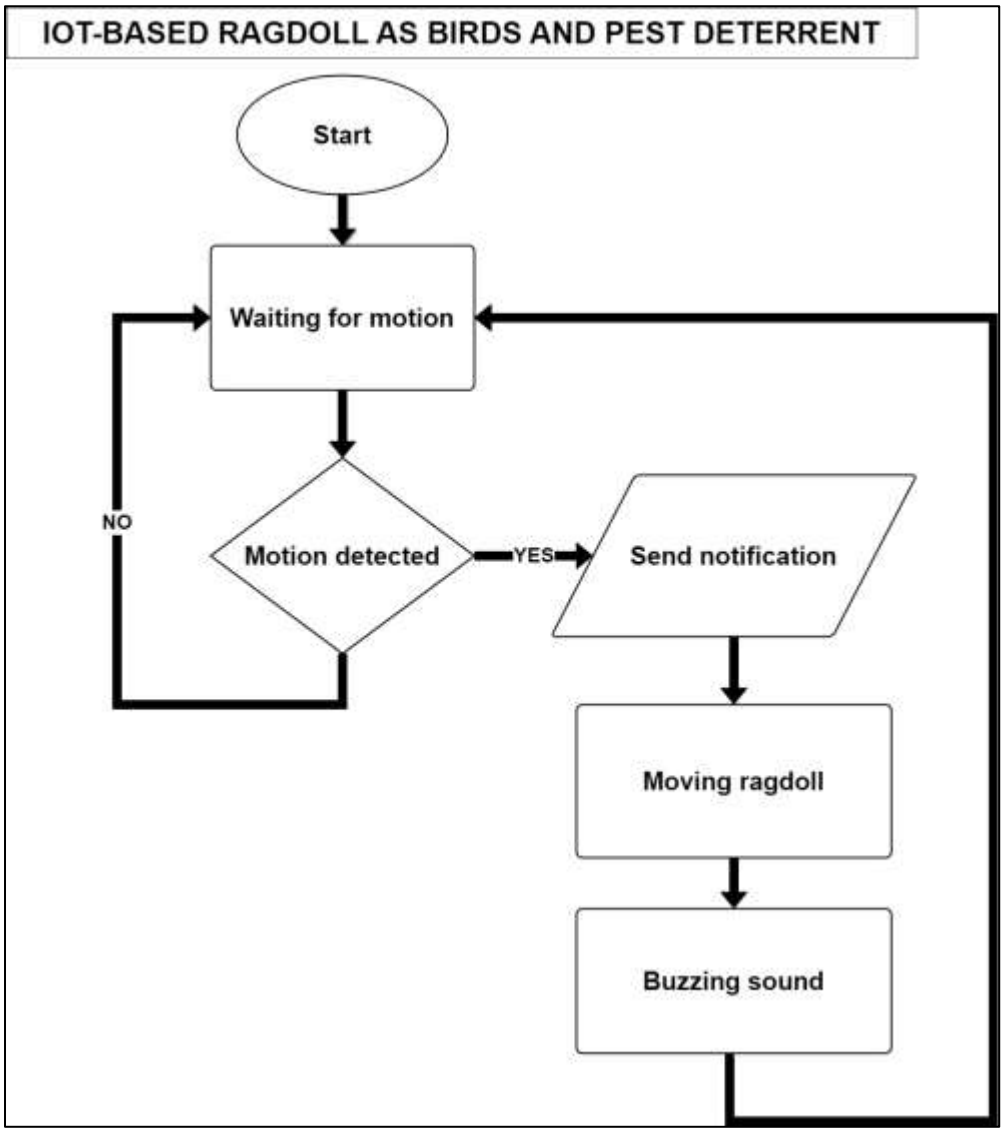

Figure 5. IoT- Based Ragdoll as Birds and Pest Deterrent System Flow Chart

Development System Process

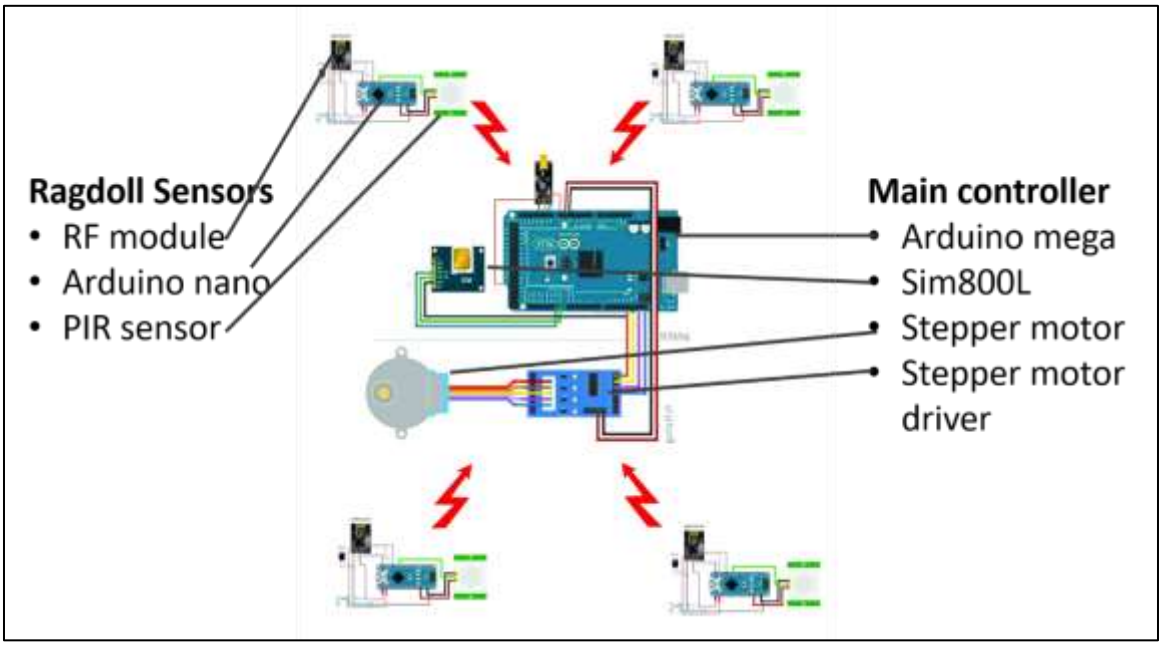

Figure 6. Development System Process

The main controller and the purpose of the RF module, stepper motor, stepper motor driver, SIM800L, and Arduino Mega in the Ragdoll system were to provide wireless communication, motor control, and text message notification capabilities. The RF (radio frequency) module is a wireless communication module that allowed the Arduino Mega to transmit and receive data wireless with the sensors of the Ragdoll system. The stepper motor driver is a module that was used to drive the stepper motor. It took input signals from the Arduino Mega and translated them into the electrical pulses that were required to control the

stepper motor. In the ragdoll system, stepper motors were responsible for simulating the movement of the ragdoll. The SIM800L is a module that provided

cellular communication capabilities, allowing the Ragdoll system to send and receive text messages. This was used for notification alerts. In summary, the RF module provides wireless communication capabilities, the stepper motor and stepper motor driver allow for precise control of movement, the SIM800L enabled text message notification, and the Arduino Mega served as the control center of the ragdoll system, coordinating communication and controlling movement.

The ragdoll sensor also played a significant role in detecting the birds. The PIR (passive infrared) sensor detects the change in infrared radiation when a bird flied within the range of the sensor. The Arduino Nano is a micro controller board that would be programmed to read the signal from the PIR sensor and send a wireless signal using the RF module into the main controller. The RF (radio frequency) module is a wireless communication module that allowed the Arduino Nano to transmit data wireless to the main controller of the Ragdoll system.

Research Locale

The study was conducted in Sagbayan, San Miguel, Surigao del Sur, Philippines, in collaboration with the farmers in the barangay as the respondents. The researchers chose Barangay Sagbayan because the farmland in this barangay is massive. Due to massive farmland, the number of birds and insects were great in this area. The manual process of deterring insects and birds is a hassle due to their inability to act on their own without the supervision of a human.

Hence, this automated ragdoll would help the farmers make their farming easier. The design of the bird repellent device worked by using computer vision techniques and PIR sensors to detect the thermal of the birds and pest infrared of each object in each frame. A micro controller then processed the data. The micro controller used sound frequency to activate the actuator once the object has been detected by the PIR. Also, with the SMS notification, the owner would be notified that there were birds, insects, and any living things that crawled in the area within a 5–7-meter radius.

Figure 7. Research Locale of the Study

Sampling Procedure

This study utilized convenience random sampling to gather data from the respondents. Convenience sampling is probably the most common of all sampling techniques. With convenience sampling, the samples are selected because they were accessible to the researcher.

Table 1. Distribution of Respondents

Population in Barangay Sagbayan, San Miguel Surigao Del Sur	Population Sample Size
Purok 1	45
Purok 2	40
Purok 3	32
Purok 4	30
Purok 5	35
Purok 6	33
Purok 7	28
Total	**243**

Sample

The respondents to this study were the farmers of the Municipality of San Miguel, specifically in Barangay Sagbayan. This study focused on those rice field owners and tenants who were currently using or practicing manual processes for deterring birds and pests around rice yield. The researchers gathered the range population by purok in Barangay Sagbayan. The total targets of this study are 243. The researchers used convenience random sampling because the samples were selected and accessible to them. Out of a 243-person sample population, the researchers identified 100 farmers as respondents to the study. The researchers chose this study to create prototypes based on the Internet of Things for monitoring and automatically controlling bird attacks to maximize crop yields in agriculture.

Sampling Techniques

Purposive sampling is a non-probability sampling method that occurs when elements selected for the sample are chosen by the researcher's judgment. Researchers often believed that they could obtain a representative sample by using sound judgment, which resulted in saving time and money. In Black (2010), the study said that, to some extent, researchers assumed to be "representatives"

of that role, although implementation studies rarely explained the rationale for selecting only some and not all of the available role representatives.

Research Instrument

The researchers used ISO 25010 as a guide questionnaire to better fit the purpose of their study as well as gathered data specifically from the farmers of Barangay Sagbayan, San Miguel. The researchers provided questions that came from the statement of the problem and were answered by the objectives of the study. An adapting questionnaire that optimized the number of propositions presented to each respondent and generates an individual learner profile based However, a survey questionnaire was conducted offline and online with a corresponding video presentation of the system as a method to evaluate the understanding of original and adapted questionnaire items to be applied in a future study.

Data Gathering Procedure

In this phase, the researchers followed several steps in gathering the data needed for this study. Subsequently, the researchers followed the procedures that would utilize: The researchers prepared a permission letter for the barangay. This letter was presented to the **Barangay Captain, Hon. Ruel A. Taladoc,** to get his approval for the conduct of the study. After the permission letter, the researchers prepared another letter to the respondents of Barangay Sagbayan, San Miguel Surigao del Sur, for their permission to participate in the said study. One of the procedures for collecting the data is the adaptation of the survey questionnaire since the researchers use a quantitative research design. Once the data was collected, the researchers tabulated the gathered data from the respondents for the analysis and interpretation. There was be a series of consultations with the adviser and the IT experts to ensure the accuracy and reliability of the embedded system.

Statistical Treatment of Data

The data collected were subjected to statistical treatment with the help of the researcher's statistician. To interpret the data effectively, the researchers utilized the non-probability sampling technique which was Convenience sampling since it is the easiest method of sampling because participants are selected based on availability and willingness to take part.

The study used Quantitative methods of inquiry. In the quantitative approach, the Average Weighted Mean was used to determine the assessment of the respondents with regard to system consistency, learnability, user guidance, system capability, efficiency, usefulness, and satisfaction. The formula is:

X= Fx N where,

x is the weighted mean

F is the frequency

X is the weight of each item

N is the number of cases

Materials

Software must be compatible with the operating system and supported the programming language to be used. The software and hardware requirements and specifications are shown in Tables 1 and 2 below.

Table 2. Software Requirements

Components	Purpose
C++	Language use in creating the system as well to design for the real-time effect of the IoT-Based Birds and Pests Deterrent system

Table 3. Hardware Requirements

Components	Specification
Arduino Mega	This device is the main controller of the system, responsible for moving the ragdoll if any bird or visible object that has a heat signature in their body in motion, it also controls the notification of the system to the client as well as receiving a signal from the sensor.
Arduino Nano	This device is the sub controller of the system that only caters to the PIR motion sensors that are

	installed 5 meters away from the main controller.
PIR Sensor	This device is responsible for detecting a bird in motion at a 7 meters radius.
SIM800L	This device is used for sending SMS notification to the client as well as missed Call Notification, it contains a sim card.
Stepper Motor	This device is for moving the ragdoll.

Schedule and Timeline

Table 4. Gantt chart

ACTIVITY	**D U R A T I O N MONTH 2022-0223**									
	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May
Title/Concept Formulation										
Chapter 1 – 3 Preparation										
Pre-Oral Defense										
Designing & coding the system										
Writing of Chapters 4 & 5										
System Evaluation										
System Testing										

Final Documentation										
Final Defense										

Project Teams and Member's Responsibilities

Every member of the team has a responsibility for the system to be successfully made. The members were given a clear task for them to do their part, contributing and sharing ideas to the group. The details included making system design and coding, gathering data. All tasks were given individually to achieve the goal of the study.

Table 5. Team Responsibilities

RESEARCHERS	RESPONSIBILITY
Jomar F. Godinez	• Programmer • Data provider • Documentation In-charge • Chapter III In-charge • System In-charge
Jeicer Jhon Gonjoran	• Documentation In-charge • Programmer • PPT in-charge • Runner • Chapter I In-charge • System In-charge
Lea Mae V. Linaza	• Encoder • Chapter I In-charge • Documentation In-charge
Jenjen M. Maca	• Encoder • Documentation In-charge

	• Chapter I In-charge • Printing In-charge
Jerome Macheca	• Data provider • Documentation In-charge • PPT in charge • Chapter III In-charge • PPT In-charge • CV In-charge
Rialyn N. Magallanes	• Encoder • Treasurer • Documentation In-charge • PPT In-charge • RRL In-charge • Documentation Revision • Chapter 4 and 5 In-charge • Letters In-charge

Budget Cost Management Plan

The table below shows the budget expenses of the system development used as time progresses and as well as the cost incurred.

Table 6. Budget Cost Management Plan (Proposed)

Quantity	Unit	Description	Unit Price	Amount
7	Pcs	Nrf24101	₱ 135.00	₱ 945.00
7	Pcs	PIR Sensor	₱ 100.00	₱ 700.00
1	Pc	Arduino Mega	₱ 850.00	₱ 850.00
7	Pcs	Arduino Nano	₱ 292.00	₱ 2,044.00
1	Pc	Sim800L	₱360.00	₱ 360.00
2	Pcs	SD-card Reader Module	₱40.00	₱ 80.00
2	Pcs	Battery	₱210.00	₱420.00

2	Pcs	Battery Cord	₱35.00	₱70.00
			Total:	**₱ 5,549.00**

Chapter IV

RESULTS AND DISCUSSION

This chapter presents the step-by-step process of how to use the system or innovation. It also presents the design, development, and results of the system evaluation.

Statistical Analysis

Scale	Range	Verbal Description
5	4.20 – 5.00	Extremely Satisfied
4	3.40 – 4.19	Very Satisfied
3	2.60 – 3.39	Satisfied
2	1.80 – 2. 59	Slightly Satisfied
1	1.00 – 1.79	Not Satisfied

Table 7. Statistical Interpretation of Data using a 5-Point Scale

System Evaluation

This section presents the results of the evaluation of the system. The said evaluation was conducted to the identified farmers of Barangay Sagbayan San Miguel Surigao del Sur, to the faculty and staff and to the students of North Eastern Mindanao State University as the end-users. The survey consists of two (2) parts namely: Part 1 - Profile of Respondents, Part 2 is the Evaluation. The Part 2 which is the evaluation consists of four (4) parameters namely: Functional Suitability, System Usability, System Reliability and Performance efficiency.

The Level of Satisfaction of the System as Perceived by the Respondents

Table 8. Functional Suitability

Indicators	Weighted Mean	Adjectival Rating
1. Functional Completeness. The prototype covers all the specified tasks and user objectives.	4.68	Extremely Satisfied
2. Functional Correctness. The prototype provides the correct results with the required level of accuracy.	4.59	Extremely Satisfied
3. Functional Fitness. The prototype facilitates the fulfillment of defined tasks and goals.	4.30	Extremely Satisfied
Total Weighted Mean:	**4.52**	**Extremely Satisfied**

Table 8 shows the evaluation resulted on the functional suitability of the study. Functional completeness got highest mean of 4.68 because the respondents were extremely satisfied in terms of the completeness as it covers all the specified tasks and user objectives. (Chan et al., 2019) stated that the owner of the rice field has been making ragdolls to increase their yield because of the physical damage that hungry birds feeding on paddy fields cause. As an outcome, farmers are losing a lot of their paddy yield harvest in terms of quantity, quality, and the money they expected from their harvests.

Table 9. System Usability

Indicators	Weighted Mean	Adjectival Rating
4. Suitability Recognition. The prototype allows the user to identify if it is suitable for their needs.	4.62	Extremely Satisfied
5. Learnability. The prototype can be utilized by specific users to accomplish specific learning objectives related to learning to use the prototypes effectively, efficiently, without restriction, without risk, and with satisfaction in a specific usage context.	4.64	Extremely Satisfied
6. Operability. The prototype has features that facilitate way and control.	4.64	Extremely Satisfied
7. Accessibility. The prototype can be utilized to accomplish a specific goal by individuals with a full spectrum of traits and skills.	4.67	Extremely Satisfied
Total Weighted Mean:	**4.64**	**Extremely Satisfied**

It can be depicted from Table 9 that the evaluation resulted in the system usability. Accessibility indicator got 4.67 which implies that the respondents were extremely satisfied. That means to say that the proposed prototype can be utilized to accomplish a specific goal by individuals with a full spectrum of traits and skills. The study is accessible and hassle-free to the farmers. Confirmed to the study of (Savitha et al. 2020) the Internet of Things, abbreviated as IoT, is a technology framework that brings technologies together to execute specific actions.

Table 10. System Reliability

Indicators	Weighted Mean	Adjectival Rating
8. Maturity. The prototype meets the requirement and is reliable under normal processing.	4.69	Extremely Satisfied
9. Availability. The prototype is functional and accessible when needed.	4.56	Extremely Satisfied
10. Fault Tolerance. The prototype works as intended despite having hardware or software errors.	4.60	Extremely Satisfied
Total Weighted Mean:	**4.61**	**Extremely Satisfied**

Table 10 ratified the evaluation result on the system reliability of the study, wherein the indicator maturity got 4.69 which implies that the respondents were extremely satisfied. That means the proposed prototype met the requirements and is reliable under normal processing. Hence, Terry et al. (2021) developed a wildlife monitoring system that utilizes passive infrared (PIR) sensors to detect the presence of wildlife within the sensor's field of view. The system was tested and evaluated in a wildlife reserve and was able to detect the activity of various wildlife species, including birds, rodents, and larger mammals.

Table 11. Performance Efficiency

Indicators	Weighted Mean	Adjectival Rating
11. Time Behavior. The prototype's response and processing times and quantity rates when performing its functions, meet the requirements.	4.67	Extremely Satisfied
12. Resource Utilization. The prototype value and types of resources used when performing its functions, meet the requirements	4.65	Extremely Satisfied
13. Capacity. The prototypes maximum limits of parameter, meet requirements.	4.70	Extremely Satisfied
Total Weighted Mean:	**4.67**	**Extremely Satisfied**

It can be seen from Table 11 the results of Performance Efficiency of the study, wherein the indicator, capacity got 4.70 adjectival rating of extremely satisfied that implies the prototype maximum limit of the parameter meets its requirements. (Gaur et al., 2021), affirmed that to develop a novel methodology to solve such challenges, the diversity and complexity of data sources also impose new trust challenges in the data collection process.

Table 12. Overall Weighted Mean

Parameters	Weighted Mean	Adjectival Rating
Functional Suitability	4:52	Extremely Satisfied
System Usability	4.64	Extremely Satisfied
System Reliability	4.61	Extremely Satisfied
Performance Efficiency	4.67	Extremely Satisfied
Overall Weighted Mean:	**4.61**	**Extremely Satisfied**

The framework ease-of-use test for "IoT-Based Ragdoll as a Bird and Pest Deterrent" is shown in Table 12. With an adjectival rating of "extremely satisfied" for respondents who agreed that the framework was dependable and simple to use, the performance efficiency received the most significant weighted midpoints of 4.67, which implies that the proposed prototypes efficiency meets all the requirements and this study is very relevant to the identified farmers of Barangay Sagbayan, San Miguel Surigao del Sur.

Last but not least, the grand weighted mean of 4.61 and the adjectival rating of "Extremely Satisfied" in the overall system test result indicate that a system is fully functional once utilized.

Chapter V

SUMMARY OF FINDINGS, CONCLUSION, AND RECOMMENDATION

This chapter presents the research work undertaken, conclusions drawn, and recommendations made as an outgrowth of this study. This study is on the profile and performance evaluation of the IoT-Based ragdoll as Birds and Pest Deterrent

Summary of Findings

The proposed prototype might be useful to farmers in terms of farming, according to the findings of the researchers. Farmers can use the prototype that has been proposed. Despite its high price, it is useful for repelling birds and other pests. The cultivating technique will be simple and have fewer problems when the ranchers utilize the computerized ragdoll. In terms of preventing birds and other pests, the system is capable of providing precise monitoring.

Additionally, the majority of the farmers in Brgy. Sagbayan gave remarks on the review. The respondents said that our system is one of a kind and that farmers can use the proposed prototype, especially during harvest time. Some of them said that our framework would be better if we worked together with the branch of agribusiness, particularly on monetary issues. The respondents also stated that our system is useful despite its high cost.

Moreover, according to the ICT and faculty personnel based on their evaluation of the researchers' deployments, the people behind it said that the proposed prototype that we proposed is feasible and working, but they suggest that we can recommend to future searchers the Raspberry Pi for more features.

Conclusion

Based on the evaluation results utilizing ISO 25010 standards criteria, it can be determined that the system performed well according to the desired requirements. The system's strong results in the assessment and performance compliance test demonstrate that it has the capability to solve the problem of the rice field owners. This prototype IoT-Based Ragdoll as Birds and Pest Deterrent is a hassle-free and great innovation regarding in the traditional ragdoll.

Recommendation

Based on the foregoing findings and conclusions, the following recommendations were offered:

- The researchers recommend this study to the farmers of Barangay Sagbayan and those farmlands that could afford this system. With the collaboration of the Municipality of Agriculture. It really helps the farmers because it is hassle-free, though it may be costly.
- For future researchers, this research would serve as their guide, and they could also update this study into something innovative. The researchers recommend that they would adapt new features; in SMS notification they

  can innovate the text message into more specified message that is applicable to the farmers. In power supply they can used solar panel. In classifying the birds and pests they can used the Raspberry Pi for new features. Researchers didn't use the Raspberry Pi due to its cost, but the devices we used were feasible and proved that the function worked well.
- Researchers recommend that they can make the system made in water proof in enhancing this study

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APPENDECIS

APPENDIX A

System Design/Assembly

Figure 8. Assembly of the IoT-Based Ragdoll as Birds and Pest Deterrent

Figure 8 shows the prototype of the IoT-Based Ragdoll as Birds and Pests Deterrent and it consist of different parts such as PIR Sensors and the Ragdoll.

Figure 9. Stepper Motor of the IoT-Based Ragdoll as Birds and Pests Deterrent

The stepper engine driver is a module that is utilized to drive the stepper engine as shown in Figure 9. It transforms the Arduino Mega's input signals into the electrical pulses necessary to control the stepper motor. In the ragdoll framework, stepper engines are capable of mimicking the development of the ragdoll.

Figure 10a. Inside of Arduino Mega Controller **Figure 10b.** Outside of Arduino Mega Controller of the IoT-Based Ragdoll as Birds and Pest Deterrent

Figure 10a shows the ragdoll system's main controller, RF module, stepper motor, stepper motor driver, SIM800L, and Arduino Mega all serve the same purpose: to enable wireless communication, motor control, and text message notification. The RF (radio frequency) module is a wireless communication module that enables the Ragdoll system's sensors and the Arduino Mega to wireless transmit and receive data. Figure 10b depicts the cover or the outside view of the Arduino Mega, the switch button and the wireless specification.

Figure 11a. Inside of Arduino Nano PIR Sensor Controller

Figure 11b. Outside of Arduino Nano PIR Sensor Controller

of the IoT-Based Ragdoll as Birds and Pest Deterrent

A micro controller board called the Arduino Nano will be programmed to read the signal from the PIR sensor and use the RF module to send a wireless signal to the main controller, as depicted in Figure 11a. The RF (radio frequency) module is a wireless communication module that enables the Arduino Nano to wireless transmit data to the Ragdoll system's primary controller. While Figure 11b shows the outside cover of the PIR sensor, it is used to cover the inside materials of the Arduino Nano to make it safe.

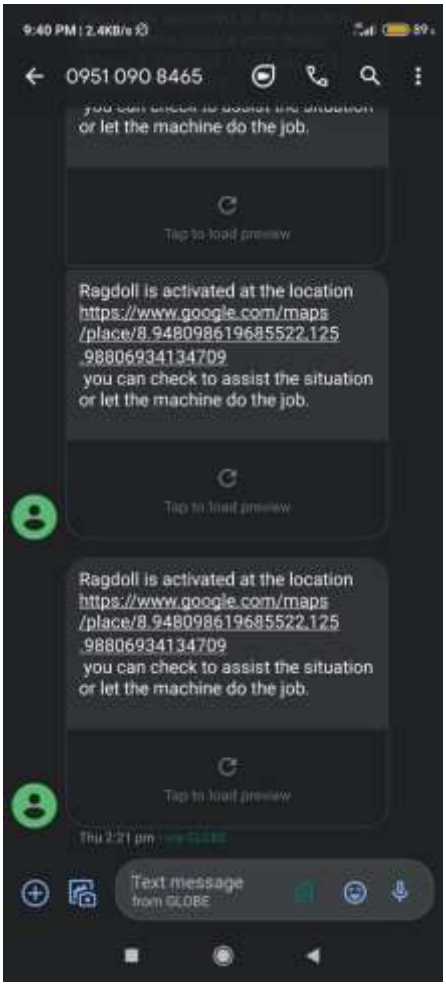


Figure 12. Sample SMS Notification

Figure 12 depicts the sample text messages received by the owner or tenant of the rice field to inform them if the ragdoll detected any objects that had an infrared temperature, and the owner or tenant would know that the ragdoll was activated. This SMS notification is one of the features of this proposed invention.

APPENDIX B

LETTER REQUEST FOR ADVISER

Republic of the Philippines
North Eastern Mindanao State University
Rosario, Tandag City, Surigao del Sur 8300
Telefax No. 086-214-4221
Website: www.nemsu.edu.ph
College Of Information Technology Education

December 12, 2022

MICHAEL L. ESTAL, MSCS
CITE Faculty
North Eastern Mindanao State University
Rosario Tandag City, Surigao del Sur

Sir:

We are 4th year students of the College of Computer of Information Technology Education of North Eastern Mindanao State University. We are currently enrolled in Special project or Thesis Writing 1. We are writing to express our sincere desire to request your assistance and expertise to serve as our Advisor on our thesis **entitled "IOT-Based Ragdoll as Birds and Pests Deterrent"** This is in partial fulfillment of the requirement for the degree of Bachelor of Science in Computer Science (BSCS).

We greatly believe that your knowledge will enrich our research work. Your dedication to your career is very valuable to us.

Your approval to this request will be greatly appreciated. Thank you very much.

GODINEZ, JOMAR F. BSCS Candidate	**GONJORAN, JEICER JHON D.** BSCS Candidate	**LINAZA, LEA MAE V.** BSCS Candidate
MACA, JEN-JEN M. BSCS Candidate	**MACHECA, JEROME P.** BSCS Candidate	**MAGALLANES, RIALYN N.** BSCS Candidate

Noted by:

CATHERINE R. ALIMBOYONG, DIT
Thesis Instructor

Acknowledged/Approved:

MICHAEL E. ESTAL, MSCS
Adviser

LETTER REQUEST FOR BARANGAY CAPTAIN OF SAGBAYAN, SAN MIGUEL

Republic of the Philippines
North Eastern Mindanao State University
Rosario, Tandag City, Surigao del Sur 8300
Telefax No. 086-214-4221
Website: www.nemsu.edu.ph
College of Information Technology Education

December 09, 2022

HON. RUEL A. TALADOC
Barangay Captain
Sagbayan, San Miguel Surigao Del Sur

Greetings!

We are 4th year students of the College of Computer of Information Technology Education of North Eastern Mindanao State University. We are currently working on our thesis **entitled "IOT-Based Ragdoll Birds and Pests Deterrent"** This is in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science (BSCS).

Anent to this, we would like to ask favor from your office to ask for the data of the number of the registered farmers by Purok in Barangay Sagbayan, San Miguel Surigao Del Sur in your office dated January 2022 – December 2022. Rest assured that the data is confidential and will be used for academic purposes only.

Your contribution for the success of this endeavor is greatly appreciated.

Thank you very much.

GODINEZ, JOMAR F.
BSCS Candidate

GONJORAN, JEICER JHON D.
BSCS Candidate

LINAZA, LEA MAE V.
BSCS Candidate

MACA, JEN-JEN M.
BSCS Candidate

MACHECA, JEROME P.
BSCS Candidate

MAGAULANES, RIALYN N.
BSCS Candidate

Noted by:

MICHAEL L. ESTAL
Research Adviser

Approved:

HON. RUEL A. TALADOC
Barangay Captain

LETTER REQUEST FOR THE MUNICIPAL AGRICULTURIST OF SAN MIGUEL

Republic of the Philippines
North Eastern Mindanao State University
Rosario, Tandag City, Surigao del Sur 8300
Telefax No. 086-214-4221
Website: www.nemsu.edu.ph
College of Information Technology Education

February 21, 2023

Sir:

We are 4th year students of the College of Computer of Information Technology Education of North Eastern Mindanao State University. We are currently working on our thesis **entitled "IOT-Based Ragdoll as Birds and Pests Deterrent"** This is in partial fulfillment of the requirement for the degree of Bachelor of Science in Computer Science (BSCS).

Anent to this, we would like to ask favor from your good office to conduct an interview regarding on how technology affect the agricultural farming. Rest assured that the data is confidential and will be used for academic purposes only.

Your contribution for the success of this endeavor is greatly appreciated.

Thank you very much.

GODINEZ, JOMAR F. BSCS Candidate	**GONJORAN, JEICER JHON D.** BSCS Candidate	**LINAZA, LEA MAE V.** BSCS Candidate
MACA, JEN-JEN M. BSCS Candidate	**MACHECA, JEROME P.** BSCS Candidate	**MAGALLANES, RIALYN N.** BSCS Candidate

Received by: Ms. Mae G.
2-7

LETTER REQUEST FOR THE CITY AGRICULTURIST OF TANDAG

Republic of the Philippines
North Eastern Mindanao State University
Rosario, Tandag City, Surigao del Sur 8300
Telefax No. 086-214-4221
Website: www.nemsu.edu.ph
College of Information Technology Education

February 16, 2023

HON. ROXANNE C. PIMENTEL
City Mayor
Tandag City, Surigao del Sur

Thru: VON C. FALCON
City Agriculturist
Awasian, Tandag City Surigao del Sur

Sir:

We are 4th year students of the College of Computer of Information Technology Education of North Eastern Mindanao State University. We are currently working on our thesis **entitled "IOT-Based Ragdoll as Birds and Pests Deterrent"** This is in partial fulfillment of the requirement for the degree of Bachelor of Science in Computer Science (BSCS).

Anent to this, we would like to ask favor from your good office to conduct an interview regarding on how technology affect the agricultural farming. Rest assured that the data is confidential and will be used for academic purposes only.

Your contribution for the success of this endeavor is greatly appreciated.

Thank you very much.

GODINEZ, JOMAR F.
BSCS Candidate

GONJORAN, JEICER JHON D.
BSCS Candidate

LINAZA, LEA MAE V.
BSCS Candidate

MACA, JENJEN M.
BSCS Candidate

MACHECA, JEROME P.
BSCS Candidate

MAGALLANES, RALYN N.
BSCS Candidate

Noted by:

MICHAEL L. ESTAL
Research Adviser

Approved:

VON C. FALCON
City Agriculturist

LETTER REQUEST FOR THE RESPONDENTS

Republic of the Philippines
North Eastern Mindanao State University
Rosario, Tandag City, Surigao del Sur 8300
Telefax No. 086-214-4221
Website: www.nemsu.edu.ph

College Of Information Technology Education

April 05, 2023

Dear Respondents:

We are 4th year students of the College of Computer of Information Technology Education of North Eastern Mindanao State University. We are currently enrolled in Special project or Thesis Writing 1. We are writing to express our sincere desire to request your assistance and expertise to serve as our Respondents on our thesis **entitled "IOT-Based Ragdoll as Birds and Pests Deterrent"** This is in partial fulfillment of the requirement for the degree of Bachelor of Science in Computer Science (BSCS).

Your approval to this request will be greatly appreciated. Thank you very much.

GODINEZ, JOMAR F. BSCS Candidate	GONJORAN, JEICER JHON D. BSCS Candidate	LINAZA, JEA MAE V. BSCS Candidate
MACA, JEN-JEN M. BSCS Candidate	MACHECA, JEROME P. BSCS Candidate	MAGALLANES, RIALYN N. BSCS Candidate

Noted by:

CATHERINE R. ALIMBOYONG, DIT
Thesis Instructor

Acknowledged/Approved:

MICHAEL E. ESTAL, MSCS
Adviser

LETTER REQUEST FOR STATISTICIAN

Republic of the Philippines
North Eastern Mindanao State University
Rosario, Tandag City, Surigao del Sur 8300
Telefax No. 086-214-4221
Website: www.nemsu.edu.ph
College Of Information Technology Education

May 08, 2023

ENGR. ALEX S. LADAGA, PhD
CET, Dean
North Eastern Mindanao State University
Rosario Tandag City, Surigao del Sur

Sir:

We are 4th year students of the College of Computer of Information Technology Education of North Eastern Mindanao State University. We are currently enrolled in Special project or Thesis Writing II. We are writing to express our sincere desire to request your assistance and expertise to serve as our Statistician on our thesis **entitled "IOT-Based Ragdoll as Birds and Pests Deterrent"** This is in partial fulfillment of the requirement for the degree of Bachelor of Science in Computer Science (BSCS).

We greatly believe that your knowledge will enrich our research work. Your dedication to your career is very valuable to us.

Your approval to this request will be greatly appreciated. Thank you very much.

GODINEZ, JOMAR F. BSCS Candidate	**GONJORAN, JEICER JHON D.** BSCS Candidate	**LINAZA, LEA MAE V.** BSCS Candidate
MACA, JEN-JEN M. BSCS Candidate	**MACHECA, JEROME P.** BSCS Candidate	**MAGALLANES, RIALYN N.** BSCS Candidate

Noted by:

MICHAEL E. ESTAL, MSCS
Thesis Adviser

CATHERINE R. ALIMBOYONG, DIT
Thesis Instructor

Acknowledged/Approved:

ALEX S. LADAGA, PhD
Statistician

LETTER REQUEST FOR IT CONSULTANT

Republic of the Philippines
North Eastern Mindanao State University
Rosario, Tandag City, Surigao del Sur 8300
Telefax No. 086-214-4221
Website: www.nemsu.edu.ph
College Of Information Technology Education

May 08, 2023

MR. MERRIT GUARTE
IT Consultant
Quezon Street Cor. Martinez Street.
Bag-ong Lungsod Tandag City, Surigao del Sur

Sir:

We are 4th year students of the College of Computer of Information Technology Education of North Eastern Mindanao State University. We are currently enrolled in Special project or Thesis Writing II. We are writing to express our sincere desire to request your assistance and expertise to serve as our IT Consultant on our thesis **entitled "IOT-Based Ragdoll as Birds and Pests Deterrent"** This is in partial fulfillment of the requirement for the degree of Bachelor of Science in Computer Science (BSCS).

We greatly believe that your knowledge will enrich our research work. Your dedication to your career is very valuable to us.

Your approval to this request will be greatly appreciated. Thank you very much.

GODINEZ, JOMAR F. BSCS Candidate	**GONJORAN, JEICER JHON D.** BSCS Candidate	**LINAZA, LEA MAE V.** BSCS Candidate
MACA, JEN-JEN M. BSCS Candidate	**MACHECA, JEROME P.** BSCS Candidate	**MAGALLANES, RIALYN N.** BSCS Candidate

Noted by:

MICHAEL E. ESTAL, MSCS Thesis Adviser	**CATHERINE R. ALIMBOYONG, DIT** Thesis Instructor

Acknowledged/Approved:

MR. MERRIT GUARTE
IT Consultant

LETTER REQUEST FOR GRAMMARIAN

Republic of the Philippines
North Eastern Mindanao State University
Rosario, Tandag City, Surigao del Sur 8300
Telefax No. 086-214-4221
Website: www.nemsu.edu.ph
College Of Information Technology Education

May 08, 2023

HENZ ANTOINETTE YUSICO
Grammarian
Telaje, Tandag City, Surigao del Sur

Ma'am:

We are 4th year students of the College of Computer of Information Technology Education of North Eastern Mindanao State University. We are currently enrolled in Special project or Thesis Writing II. We are writing to express our sincere desire to request your assistance and expertise to serve as our Grammarian on our thesis **entitled "IOT-Based Ragdoll as Birds and Pests Deterrent"** This is in partial fulfillment of the requirement for the degree of Bachelor of Science in Computer Science (BSCS).

We greatly believe that your knowledge will enrich our research work. Your dedication to your career is very valuable to us.

Your approval to this request will be greatly appreciated. Thank you very much.

GODINEZ, JOMAR F. BSCS Candidate	**GONJORAN, JEICER JHON D.** BSCS Candidate	**LINAZA, LEA MAE V.** BSCS Candidate
MACA, JEN-JEN M. BSCS Candidate	**MACHECA, JEROME P.** BSCS Candidate	**MAGALLANES, RIALYN N.** BSCS Candidate

Noted by:

MICHAEL E. ESTAL, MSCS
Thesis Adviser

CATHERINE R. ALMBOYONG, DIT
Thesis Instructor

Acknowledged/Approved:

HENZ ANTOINETTE YUSICO
Grammarian

ENDORSEMENT LETTER OF ADVISER

Republic of the Philippines
North Eastern Mindanao State University
Rosario, Tandag City, Surigao del Sur 8300
Telefax No. 086-214-4221
www.nemsu.edu.ph

College of Information Technology Education
Department of Computer Studies

May 15, 2023

TO THE PANEL OF EVALUATORS

This is to endorse the following students:
GODINEZ, JOMAR F., GONJORAN, JEICER JOHN D., LINAZA, LEA MAE V., MACA, JEN-JEN M., MACHECA, JEROME P., MAGALLANES, RIALYN N., to present their Thesis entitled: **(IOT-BASED RAGDOLL AS BIRDS AND PEST DETERRENT)**

This is to further certify that I have thoroughly checked and reviewed their outputs prior to their final oral presentation.

MICHAEL E. ESTAL, MSCS
Full Name and Signature of Thesis Adviser

PLAGIARISM RESULT

Republic of the Philippines
North eastern Mindanao State University
Rosario, Tandag City, Surigao del Sur 8300
Telefax No. 086-214-4221
Website: www.sdssu.edu.ph

RESEARCH & PUBLICATION OFFICE

C E R T I F I C A T I O N

This is to certify that the paper entitled IOT-BASED RACDOLL AS BIRDS AND PEST DETERRENT by the author/s JONAR F. GODINES, RIALIN MAGALLANES, JEROME MACHECA, JENSEN MACA, JEKER JHON GONJORAN, LEA MAE LINAZA, has been subjected to plagiarism and grammarly checkers dated 05-15-23 with the following results:

	Originality	Grammar Score
Abstract:	100 %	85 %
Chapter 1:	83 %	80 %
Chapter 2:	90 %	81 %
Chapter 3:	97 %	83 %
Chapter 4:	98 %	80 %
Chapter 5:	95 %	93 %

This certification is issued on 05-16-23, at NEMSU, Tandag City, Surigao del Sur.

O.R. No. UNDERGRAD
Date Paid: _____

MELANI JOY S. GALLARDO
Staff

Noted:

ROLLY G. SALVALEON, PhD
Director, Research & Development

FM-R&D-005M

APPENDIX C

CERTIFICATE OF ADVISER

Republic of the Philippines
North Eastern Mindanao State University
Rosario, Tandag City, Surigao del Sur 8300
Telefax No. 086-211-4221
www.nemsu.edu.ph

College of Information Technology Education
Department of Computer Studies

CERTIFICATION

I certify that I have provided suggestions, comments and that I have checked and evaluated the manuscript of my advisees, with paper titled:

IOT-BASED RAGDOLL AS BIRDS AND PESTS DETERRENT

MICHAEL L. ESTAL, MSCS
Name and Signature of Thesis Adviser
Date December 12, 2022

CERTIFICATE OF BRGY. CAPTAIN OF SAGBAYAN SAN MIGUEL SDS

Republic of the Philippines
North Eastern Mindanao State University
Rosario, Tandag City, Surigao del Sur 8300
Telefax No. 086-214-4221
Website: www.nemsu.edu.ph
College Of Information Technology Education

CERTIFICATION

To Whom It May Concern:

This is to certify that the study entitled **"IOT-BASED RAGDOLL AS BIRDS AND PEST DETERRENT"** has been approved and allowed to conduct their study in Barangay Sagbayan, San Miguel Surigao del Sur by the undersigned.

This certification is issued upon the request of the researchers. Given this 21st of May 2023 at North Eastern Mindanao State University, Main Campus, Tandag City.

HON. RUEL A. TALADOC
Barangay Captain

CERTIFICATE OF STATISTICIAN

Republic of the Philippines
North Eastern Mindanao State University
Rosario, Tandag City, Surigao del Sur 8300
Telefax No. 086-211-4221
www.nemsu.edu.ph

College of Information Technology Education
Department of Computer Studies

CERTIFICATION

To Whom It May Concern:

This is to certify that the study entitled **"IOT-BASED RAGDOLL AS BIRDS AND PEST DETERRENT"** has been reviewed and statistically checked by the undersigned. This certification is issued upon the request of the researchers.

Given this 8th of May 2023 at North Eastern Mindanao State University, Main Campus, Tandag City.

ENGR. ALEX S. LADAGA
Statistician

CERTIFICATE OF IT CONSULTANT

Republic of the Philippines
North Eastern Mindanao State University
Rosario, Tandag City, Surigao del Sur 8300
Telefax No. 086-211-4221
www.nemsu.edu.ph

College of Information Technology Education
Department of Computer Studies

CERTIFICATION

To Whom It May Concern:

This is to certify that the study entitled **"IOT-BASED RAGDOLL AS BIRDS AND PEST DETERRENT"** has been reviewed and checked the hardware requirements of this project This certification is issued upon the request of the researchers.

Given this 9th of MAY 2023 at North Eastern Mindanao State University, Main Campus, Tandag City.

MR. MERRIT GUARTE
IT Consultant

CERTIFICATE OF GRAMMARIAN

Republic of the Philippines
North Eastern Mindanao State University
Rosario, Tandag City, Surigao del Sur 8300
Telefax No. 086-211-4221
www.nemsu.edu.ph

College of Information Technology Education
Department of Computer Studies

CERTIFICATION

To Whom It May Concern:

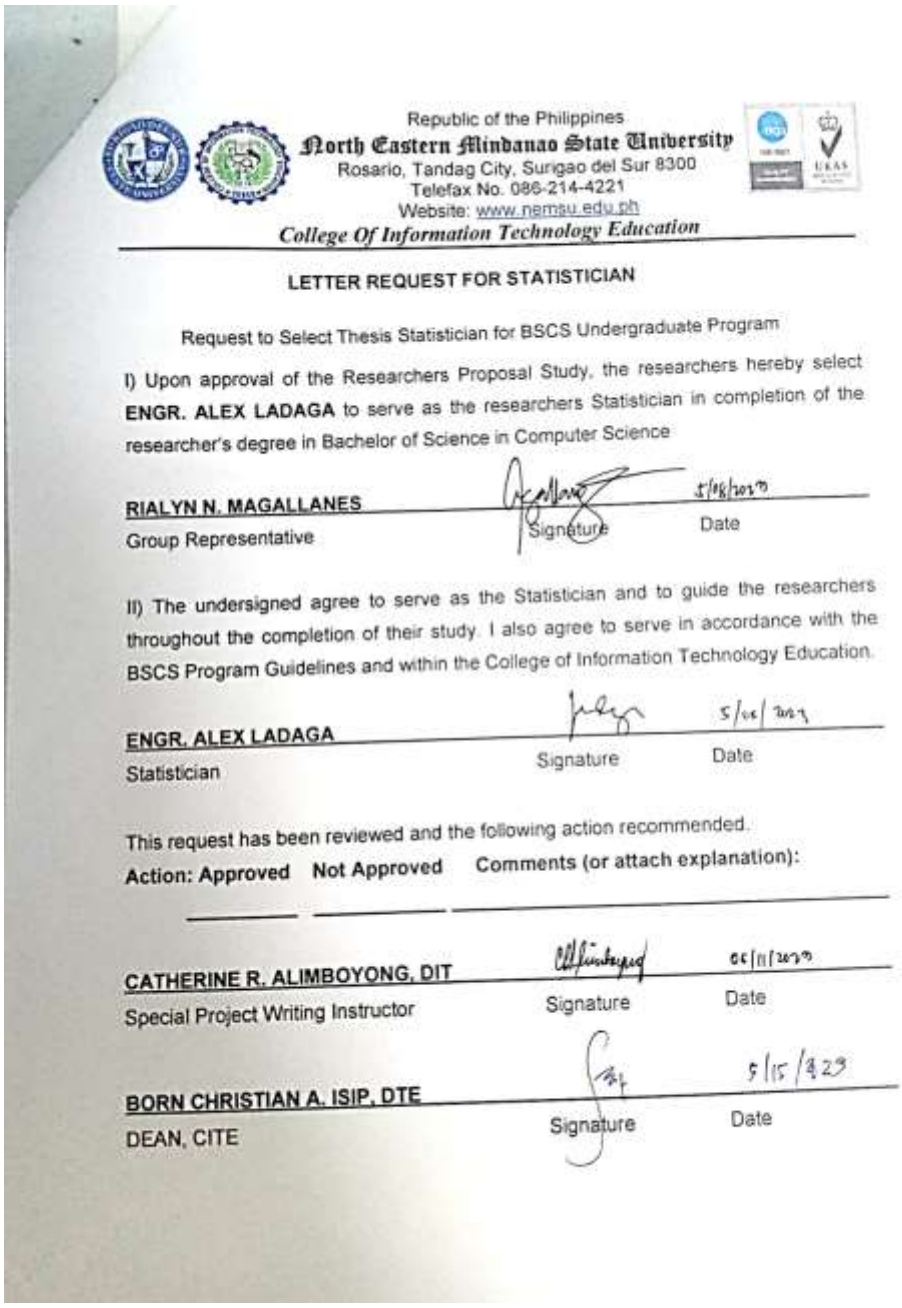
This is to certify that the study entitled **"IOT-BASED RAGDOLL AS BIRDS AND PEST DETERRENT"** has been reviewed and edited by the undersigned. This certification is issued upon the request of the researchers.

Given this 10th of MAY, 2023 at North Eastern Mindanao State University, Main Campus, Tandag City.

MS. HENZ ANTOINETTE YUSICO
Grammarian

APPENDIX D

APPROVED LETTER REQUEST FOR STATISTICIAN



Republic of the Philippines
North Eastern Mindanao State University
Rosario, Tandag City, Surigao del Sur 8300
Telefax No. 086-214-4221
Website: www.nemsu.edu.ph
College Of Information Technology Education

LETTER REQUEST FOR STATISTICIAN

Request to Select Thesis Statistician for BSCS Undergraduate Program

I) Upon approval of the Researchers Proposal Study, the researchers hereby select **ENGR. ALEX LADAGA** to serve as the researchers Statistician in completion of the researcher's degree in Bachelor of Science in Computer Science

RIALYN N. MAGALLANES — Signature — 5/08/2023
Group Representative — Signature — Date

II) The undersigned agree to serve as the Statistician and to guide the researchers throughout the completion of their study. I also agree to serve in accordance with the BSCS Program Guidelines and within the College of Information Technology Education.

ENGR. ALEX LADAGA — Signature — 5/08/2023
Statistician — Signature — Date

This request has been reviewed and the following action recommended.
Action: Approved Not Approved Comments (or attach explanation):

CATHERINE R. ALIMBOYONG, DIT — Signature — 05/11/2023
Special Project Writing Instructor — Signature — Date

BORN CHRISTIAN A. ISIP, DTE — Signature — 5/15/2023
DEAN, CITE — Signature — Date

APPROVED LETTER REQUEST FOR IT CONSULTANT

Republic of the Philippines
North Eastern Mindanao State University
Rosario, Tandag City, Surigao del Sur 8300
Telefax No. 086-214-4221
Website: www.nemsu.edu.ph
College Of Information Technology Education

LETTER REQUEST FOR IT CONSULTANT

Request to Select Thesis IT Consultant for BSCS Undergraduate Program

I) Upon approval of the Researchers Proposal Study, the researchers hereby select MR. **MERRIT GUARTE** to serve as the researchers IT Consultant in completion of the researcher's degree in Bachelor of Science in Computer Science.

RIALYN N. MAGALLANES — Signature — 05/08/23
Group Representative — Signature — Date

II) The undersigned agree to serve as the IT Consultant and to guide the researchers throughout the completion of their study. I also agree to serve in accordance with the BSCS Program Guidelines and within the College of Information Technology Education.

MR. MERRIT GUARTE — Signature — 05/11/2023
IT Consultant — Signature — Date

This request has been reviewed and the following action recommended.

Action: Approved **Not Approved** **Comments (or attach explanation):**

CATHERINE R. ALIMBOYONG, DIT — Signature — 05/11/2023
Special Project Writing Instructor — Signature — Date

BORN CHRISTIAN A. ISIP, DTE — Signature — 5/15/2023
DEAN, CITE — Signature — Date

APPROVED LETTER REQUEST FOR GRAMMARIAN

Republic of the Philippines
North Eastern Mindanao State University
Rosario, Tandag City, Surigao del Sur 8300
Telefax No. 086-214-4221
Website: www.nemsu.edu.ph

College Of Information Technology Education

LETTER REQUEST FOR GRAMMARIAN

Request to Select Thesis Grammarian for BSCS Undergraduate Program

I) Upon approval of the Researchers Proposal Study, the researchers hereby select MS. **HENZ ANTOINETTE YUSICO** to serve as the researcher's Grammarian in completion of the researcher's degree in Bachelor of Science in Computer Science.

RIALYN N. MAGALLANES		05/08/23
Group Representative	Signature	Date

II) The undersigned agree to serve as the Grammarian and to guide the researchers throughout the completion of their study. I also agree to serve in accordance with the BSCS Program Guidelines and within the College of Information Technology Education.

MS. HENZ ANTOINETTE YUSICO		05/10/2023
Grammarian	Signature	Date

This request has been reviewed and the following action recommended.

Action: Approved **Not Approved** **Comments (or attach explanation):**

CATHERINE R. ALIMBOYONG, DIT		05/11/2023
Special Project Writing Instructor	Signature	Date
BORN CHRISTIAN A. ISIP, DTE		5/15/2023
DEAN, CITE	Signature	Date

APPENDIX E

Data Gathering Tool

IoT-Based Ragdoll as Birds and Pest Deterrent

Good day! We, the 4th year BSCS students of the College of Computer of Information Technology Education of North Eastern Mindanao State University, are conducting a thesis entitled "IoT-Based Ragdoll as Birds and Pests Deterrent". In line with this, we are humbly asking for your precious time and effort to answer the following questions.

CONFIDENTIALITY. All responses of each respondent will be treated with utmost confidentiality. The information will be stored securely and will only be used for research purposes. Only the researchers and their thesis adviser will have access to the information. All data gathered from the offline/online survey will not be shared with anybody outside the research team. How does the developed system comply with the ISO 25010 software quality standards in terms of the following factors:

Part I. Profile

Profile	Category	Remarks
Type of Respondent	Owner	
	Tenant	
Age (Years old)	15 – 35	
	36 – 50	
	51 and above	
Gender	Male	
	Female	
Education	Elementary Level	
	High School Level	
	College Level	

Part II. Evaluation

After operating the capabilities of the prototype. Evaluate the prototype features of the proposed study. Please rate the level by putting check mark (/) on the circle that corresponds to your assessment. Use the scale below. Numeric Value Equivalent Rating ***5 - Very satisfied 4 - Satisfied 3 - Unsure 2 - Dissatisfied 1 - Very dissatisfied***

Indicators	5	4	3	2	1
Functional Suitability					
1. Functional Completeness. The prototype covers all the specified tasks and user objectives.	○	○	○	○	○
2. Functional Correctness. The prototype provides the correct results with the required level of accuracy.	○	○	○	○	○
3. Functional Fitness. The prototype facilitates the fulfillment of defined tasks and goals.	○	○	○	○	○
System Usability					
4. Suitability Recognition. The prototype allows the user to identify if it is suitable for their needs.	○	○	○	○	○
5. Learnability. The prototype can be utilized by specific users to accomplish specific learning objectives related to learning to use the prototypes effectively, efficiently, without restriction, without risk, and with satisfaction in a specific usage context.	○	○	○	○	○
6. Operability. The prototype has features that facilitate way and control.	○	○	○	○	○
7. Accessibility. The prototype can be utilized to accomplish a specific goal by individuals with a full spectrum of traits and skills.	○	○	○	○	○
System Reliability					
8. Maturity. The prototype meets the requirement and is reliable under normal processing.	○	○	○	○	○
9. Availability. The prototype is functional and accessible when needed.	○	○	○	○	○
10. Fault Tolerance. The prototype works as intended despite having hardware or software errors.	○	○	○	○	○
Performance Efficiency					
11. Time Behavior. The prototype's response and processing times and quantity rates when performing its functions, meet the requirements.	○	○	○	○	○
12. Resource Utilization. The prototype value and types of resources used when performing its functions, meet the	○	○	○	○	○

requirements					
13.Capacity. The prototypes maximum limits of parameter, meet requirements.	◯	◯	◯	◯	◯

Do you understand the concept of the prototype?

Yes ◯ No ◯

If no, which part did you not understand?

_____

Do you encounter any difficult while using the prototype?

Yes ◯ No ◯

If yes, please state the aspect of the prototype that was difficult.

_____

Kindly recommend any improvement(s) for the prototype (if applicable)

_____

_____

Interviewee Signature

APPENDIX F

APPROVED MINUTES DURING THESIS PROPOSAL

Republic of the Philippines
North Eastern Mindanao State University
Tandag City, Surigao del Sur
Telefax No. 086-214-4221
086- 214-2723
www.sdssu.edu.ph

TITLE: IoT-Based Ragdoll as Birds and Pests Deterrent
PROPOSAL DATE: December 14, 2022
ADVISER: Michael E. Estal, MSCS
PANEL CHAIRMAN: Coravil Joy C. Avila, MIT
Members: Cherly B. Sardovia, MSCS
Markdy Y. Orong, DIT
Eltimar T. Castro, Jr., DIT

Comments/Suggestion	Panel	Remarks
1. Objectives of the study • No objectives found in the manuscript **2. Statement of the problem** • Revise or rephrase SOP 1 and 2 **3. Conceptual Framework** • Theory where the study is anchored should be reflected in the second box of your conceptual framework **4. Chapter 2** • Apply in-text citation properly in the chapter 2; use recent literatures only • Add related literature and studies as there are few studies reflected in the manuscript **5. System flowchart** • Make it readable because it is too small: revise the system flowchart	**Coravil Joy Avila, MIT**	

Republic of the Philippines
North Eastern Mindanao State University
Tandag City, Surigao del Sur
Telefax No. 086-214-4221
086- 214-2723
www.sdssu.edu.ph

1. **Conceptual Framework** • Theory where the study is anchored should be reflected in the second box of your conceptual framework 2. **Scope and Limitation of the Study** • Specify what type of pests your system will consider; please revise the scope of the study 3. **Data Gathering Procedure** • Revise the data gathering section; translate it into a paragraph form 4. **Prototype** • Give brief explanation per pictures	**Cherly Sardovia, MSCS**	
Features suggested; • Follow the fundamentals of writing the introduction; needs revision - Revise your Objectives - Recast your SOPs - Revisit your data gathering procedure, significance of study, definition of terms • RRLs should be recent; distinguish related studies and the related literature • Provide a comprehensive synthesis of your study • For the scope of the study, it is suggested that there should be proper identification of birds (either good or bad) • Specify the methods and design of the	**Dr. Catherine R. Alimboyong**	

Republic of the Philippines
North Eastern Mindanao State University
Tandag City, Surigao del Sur
Telefax No. 086-214-4221
086- 214-2723
www.sdssu.edu.ph

proposed study: describe where the study is anchored, (what theory or model) • Ensure that there will be a validated questionnaire that will be used during the project testing or implementation • Use a standardized questionnaire to do the evaluation • Establish your interview questionnaire that would form part of the data gathering: ensure that the data collected will be used in the study		
1. Research Locale • In the research locale, specify the number of rice fields where your system will be tested or implemented. **2. Scope and Limitation** • Since you focus on pest, is black bugs also consider to be detected? Since it is considered as pest with negative impact to the rice field. • In the evaluation part of the study, kindly properly specify, who will rate it's of the indicator in your evaluation. Maybe they can specify it in the scope and limitation.	**Dr. Markdy Y. Orong,**	
Features suggested; • How can you make sure that only those insects and bird which is harmful can be affected? • What are those harmful insects and good	**Dr. Eltimar T. Castro, Jr.**	

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North Eastern Mindanao State University
Tandag City, Surigao del Sur
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insects that can be affected and can't be affected? • How are you going to simulate your project? • This thesis is good but needs more supporting research and experimentation as evidence that it is working. • Not all insects are harmful, most insect helps the rice plant/ and rice field healthy. • Consult experts' relation insects and agriculture		

Republic of the Philippines
North Eastern Mindanao State University
Tandag City, Surigao del Sur
Telefax No. 086-214-4221
086- 214-2723
www.sdssu.edu.ph

Prepared by:

GODINEZ, JOMAR F. Researcher	**GONJORAN, JEICER JHON D.** Researcher	**LINAZA, LEA MAE V.** Researcher
MACA, JENJEN M. Researcher	**MACHECA, JEROME P.** Researcher	**MAGALLANES, RIALYN N.** Researcher

Attested by:

MICHAEL E. ESTAL, MSCS
Adviser

CATHERINE R. ALIMBOYONG, DIT
Thesis Instructor

Approved by:

CORAVIL JOY C. AVILA, MIT
Chair, Panel of Evaluators

CHERLY B. SARDOVIA, MSCS
Panel Member

MARKDY Y. ORONG, DIT
Panel Member

ELTIMAR T. CASTRO, JR., DIT
Panel Member

APPROVED COMPLIANCE DURING THESIS PROPOSAL

Republic of the Philippines
North Eastern Mindanao State University
Rosario, Tandag City, Surigao del Sur 8300
Telefax No. 086-214-4221
Website: www.nemsu.edu.ph
College Of Information Technology Education

TITLE: IoT-Based Ragdoll as Birds and Pests Deterrent
PROPOSAL DATE: December 14, 2022
ADVISER: Michael E. Estal, MSCS
PANEL CHAIRMAN: Coravil Joy C. Avila, MIT
Members: Cherly B. Sardovia, MSCS
Markdy Y. Orong, DIT
Eltimar T. Castro, Jr., DIT

Notations and Suggestions	*Page Number*	*Remarks*
1. Conceptual Framework • Theory where the study is anchored should be reflected in the second box of your conceptual framework	7	**Complied**
2. Statement of the problem • Revise or rephrase SOP 1 and 2	8	**Complied**
3. Objectives of the study • No objectives found in the manuscript	9	**Complied**
• **Scope and Limitation of the Study** • Specify what type of pests your system will consider; please revise the scope of the study • Since you focus on pest, is black bugs also consider to be detected? Since it is considered as pest with negative impact to the rice field. • For the scope of the study, it is suggested	9	**Compiled**

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North Eastern Mindanao State University
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Website: www.nemsu.edu.ph
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that there should be proper identification of birds (either good or bad) • How can you make sure that only those insects and bird which is harmful can be affected? • What are those harmful insects and good insects that can be affected and can't be affected? • How are you going to simulate your project? • This thesis is good but needs more supporting research and experimentation as evidence that it is working. • Not all insects are harmful, most insect helps the rice plant/ and rice field healthy		
4. Chapter 2 • Apply in-text citation properly in the chapter 2; use recent literatures only • Add related literature and studies as there are few studies reflected in the manuscript • RRLs should be recent; distinguish related studies and the related literature • Provide a comprehensive synthesis of your study	15-21	**Complied**
5. Research Design • Specify the methods and design of the proposed study: describe where the study is anchored, (what theory or model)	24	**Complied**

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6. System Flowchart • Make it readable because it is too small: revise the system flowchart.	26	**Complied**
7. Research Locale • In the research locale, specify the number of rice fields where your system will be tested or implemented.	28	**Complied**
• **Research Instrument** • In the evaluation part of the study, kindly properly specify, who will rate it's of the indicator in your evaluation. Maybe they can specify it in the scope and limitation • Ensure that there will be a validated questionnaire that will be used during the project testing or implementation • Use a standardized questionnaire to do the evaluation • Establish your interview questionnaire that would form part of the data gathering: ensure that the data collected will be used in the study • Consult experts' relation insects and agriculture	32	**Complied**
8. Data Gathering Procedure Revise the data gathering section; translate it into a paragraph form	32	**Complied**
9. Prototype		

Republic of the Philippines
North Eastern Mindanao State University
Tandag City, Surigao del Sur
Telefax No. 086-214-4221
086- 214-2723
www.sdssu.edu.ph

Prepared by:

GODINEZ, JOMAR F. Researcher	GONJORAN, JEICER JHON D. Researcher	LINAZA, LEA MAE V. Researcher
MACA, JENJEN M. Researcher	MACHECA, JEROME P. Researcher	MAGALLANES, RALYN N. Researcher

Attested by:

MICHAEL E. ESTAL, MSCS Adviser	CATHERINE R. ALIMBOYONG, DIT Thesis Instructor

Approved by:

CORAVIL JOY C. AVILA, MIT
Chair, Panel of Evaluators

CHERLY B. SARDOVIA, MSCS Panel Member	MARKDY Y. ORONG, DIT Panel Member

ELTIMAR T. CASTRO, JR., DIT
Panel Member

APPROVED MINUTES DURING THESIS FINAL DEFENSE

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North Eastern Mindanao State University
Rosario, Tandag City, Surigao del Sur 8300
Telefax No. 086-214-4221
Website: www.nemsu.edu.ph
College Of Information Technology Education

Minutes of the Final Thesis Defense
Venue: Computer Lab 1
May 16, 2023

TITLE: IoT-Based Ragdoll as Birds and Pest Deterrent
Adviser: Michael E. Estal, MSCS
Panel Chairman: Coravil Joy C. Avila, MIT
Members: Cherly B. Sardovia, MSCS
Markdy Y. Orong, DIT
Eltimar T. Castro, Jr., DIT

Comments/Suggestion	Panel	Remarks
1. Should consider the weather, is the system water proof? 2. Change the objectives in chapter 5, specifically on the summary of findings	Coravil Joy Avila, MIT	
1. If the user will text STO, there should be a corresponding message to the owner that the ragdoll has stopped beeping. 2. Objective of the Study must answer the SOP, objective of the study must be implemented on the system	Cherly Sardovia, MSCS	

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North Eastern Mindanao State University
Rosario, Tandag City, Surigao del Sur 8300
Telefax No. 086-214-4221
Website: www.nemsu.edu.ph
College Of Information Technology Education

1. Demonstrate it on the actual rice field to determine if the system can really detect the pests	**Dr. Catherine R. Alimboyong**
1. Consider the Inclusion criteria in choosing the respondents 2. Revisit your Results and Discussions	**Dr. Markdy Y. Orong,**
1. Include in the discussion the comments from the respondents 2. Revisit your Results and Discussions 3. Secure any proof or evidence of actual simulation in the field	**Dr. Eltimar T. Castro, Jr.**

Republic of the Philippines
North Eastern Mindanao State University
Rosario, Tandag City, Surigao del Sur 8300
Telefax No. 086-214-4221
Website: www.nemsu.edu.ph
College Of Information Technology Education

Prepared by:

GODINEZ, JOMAR F.
Researcher

GONJORAN, JEICER JHON D.
Researcher

LINAZA, LEA MAE V.
Researcher

MACA, JENJEN M.
Researcher

MACHECA, JEROME P.
Researcher

MAGALLANES, RIALYN N.
Researcher

Attested by:

MICHAEL E. ESTAL, MSCS
Adviser

CATHERINE R. ALIMBOYONG, DIT
Thesis Instructor

Approved by:

CORAVIL, JOY C. AVILA, MIT
Chair, Panel of Evaluators

CHERLY B. SARDOVIA, MSCS
Panel Member

MARKDY Y. ORONG, DIT
Panel Member

ELTIMAR T. CASTRO, JR., DIT
Panel Member

APPROVED COMPLIANCE DURING THESIS FINAL DEFENSE

Republic of the Philippines
North Eastern Mindanao State University
Rosario, Tandag City, Surigao del Sur 8300
Telefax No. 086-214-4221
Website: www.nemsu.edu.ph
College Of Information Technology Education

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Adviser: Michael E. Estal, MSCS
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Members: Cherly B. Sardovia, MSCS
Markdy Y. Orong, DIT
Eltimar T. Castro, Jr., DIT

Comments/Suggestion	Page Number	Remarks
1. Should consider the weather, is the system water proof? 2. Change the objectives in chapter 5, specifically on the summary of findings	48	Complied
1. If the user will text STOP, there should be a corresponding message to the owner that the ragdoll has stopped beeping. 2. Objective of the Study must answer the SOP, objective of the study must be implemented on the system	48	Complied

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Telefax No. 086-214-4221
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1. Demonstrate it on the actual rice field to determine if the system can really detect the pests		Complied
1. Consider the Inclusion criteria in choosing the respondents 2. Revisit your Results and Discussions	34	Complied
1. Include in the discussion the comments from the respondents 2. Revisit your Results and Discussions 3. Secure any proof or evidence of actual simulation in the field		Complied

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North Eastern Mindanao State University
Rosario, Tandag City, Surigao del Sur 8300
Telefax No. 086-214-4221
Website: www.nemsu.edu.ph
College Of Information Technology Education

Prepared by:

GODINEZ, JOMAR F.
Researcher

GONJORAN, JEICER JHON D.
Researcher

LINAZA, LEA MAE V.
Researcher

MACA, JENJEN M.
Researcher

MACHECA, JEROME P.
Researcher

MAGALLANES, RIALYN N.
Researcher

Attested by:

MICHAEL E. ESTAL, MSCS
Adviser

CATHERINE R. ALIMBOYONG, DIT
Thesis Instructor

Approved by:

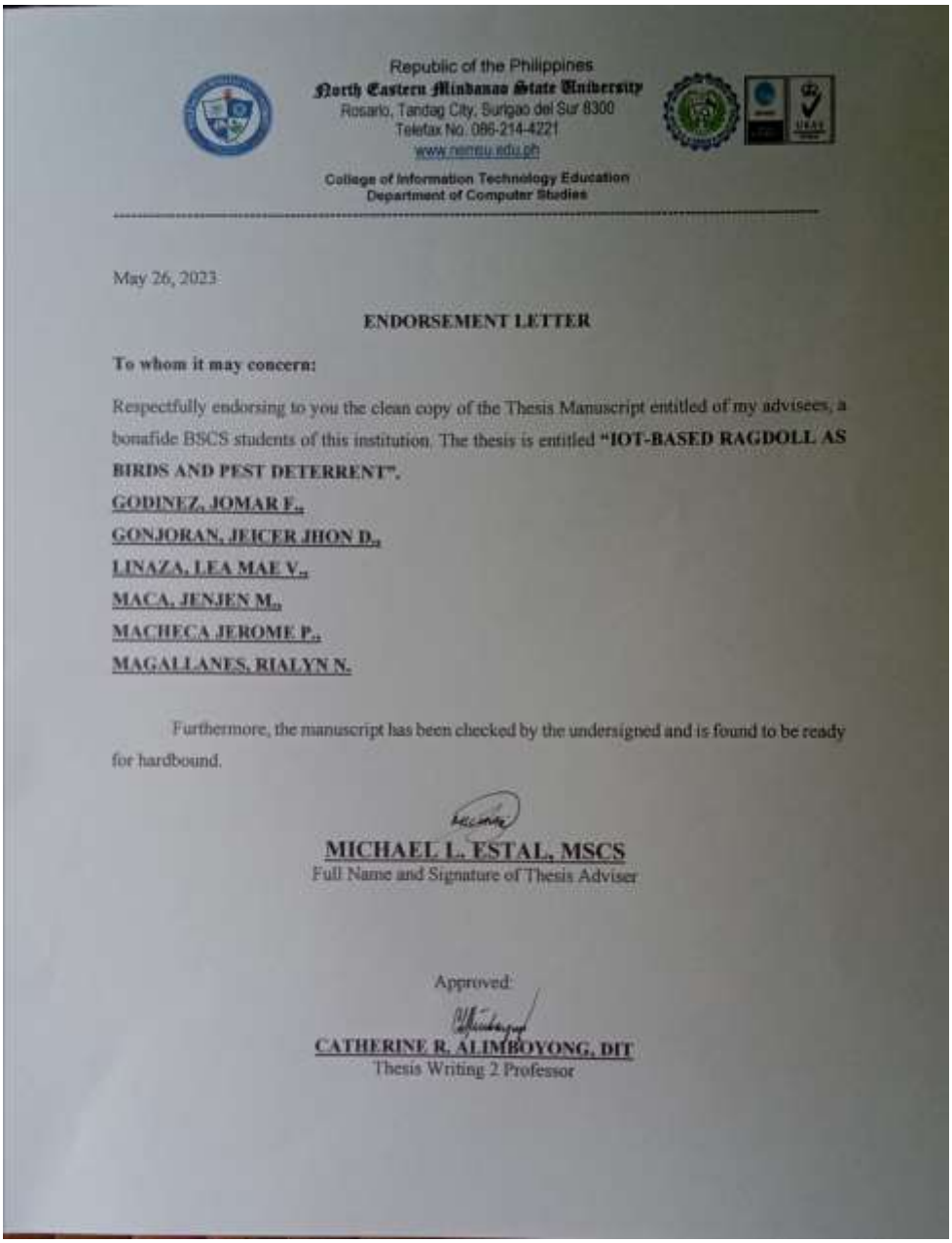
CORAVIL JOY C. AVILA, MIT
Chair, Panel of Evaluators

CHERLY B. SARDOVIA, MSCS
Panel Member

MARKDY Y. ORONG, DIT
Panel Member

ELTIMAR T. CASTRO, JR., DIT
Panel Member

APPROVED ENDORSEMENT LETTER FOR HARDBOUND



Republic of the Philippines
North Eastern Mindanao State University
Rosario, Tandag City, Surigao del Sur 8300
Telefax No. 086-214-4221
www.nemsu.edu.ph

College of Information Technology Education
Department of Computer Studies

May 26, 2023

ENDORSEMENT LETTER

To whom it may concern:

Respectfully endorsing to you the clean copy of the Thesis Manuscript entitled of my advisees, a bonafide BSCS students of this institution. The thesis is entitled **"IOT-BASED RAGDOLL AS BIRDS AND PEST DETERRENT".**

GODINEZ, JOMAR F.,
GONJORAN, JEICER JHON D.,
LINAZA, LEA MAE V.,
MACA, JENJEN M.,
MACHECA JEROME P.,
MAGALLANES, RIALYN N.

Furthermore, the manuscript has been checked by the undersigned and is found to be ready for hardbound.

MICHAEL L. ESTAL, MSCS
Full Name and Signature of Thesis Adviser

Approved:

CATHERINE R. ALIMBOYONG, DIT
Thesis Writing 2 Professor

APPENDIX G

CODES

```
#include <RF24.h>

#include <RF24Network.h>

#include <SD.h>

#include <SPI.h>

#include <SoftwareSerial.h>


SoftwareSerial sim(10, 11);

int _timeout;

String _buffer;

String number = "+639305979796"; //-> change with your number


File myFile;


int pinCS = 53; // Pin 10 on Arduino Uno



RF24 radio(7, 8);                    // nRF24L01 (CE,CSN)

RF24Network network(radio);      // Include the radio in the network

const uint16_t this_node = 01;   // Address of our node in Octal format
( 04,031, etc)


const int RELAY_PIN = 32;  // the Arduino pin, which connects to the IN pin
of relay

int x = 0;
```

```
int y = 0;
int var1 = 1;
String data;
String Bucket;
String Contact_01;
String Location;
int buzzer=40;
int LED_PIN=33;
String Parameter;
String Mess;

void setup() {
  Serial.begin(9600);
  _buffer.reserve(50);
  Serial.println("Sistem Started...");
  sim.begin(9600);
  delay(1000);
  SPI.begin();
  radio.begin();
  network.begin(90, this_node); //(channel, node address)
  pinMode(pinCS, OUTPUT);
  pinMode(buzzer, OUTPUT);
  pinMode(LED_PIN, OUTPUT);
  pinMode(RELAY_PIN, OUTPUT);
  // SD Card Initialization
```

```
  if (SD.begin())
  {
    Serial.println("SD card is ready to use.");
  } else
  {
    Serial.println("SD card initialization failed");
    digitalWrite(buzzer, HIGH);
    delay(1000);
    digitalWrite(buzzer, LOW);
    delay(1000);
    digitalWrite(buzzer, HIGH);
    delay(1000);
    digitalWrite(buzzer, LOW);
    return;
  }
digitalWrite(LED_PIN, HIGH);
getSDcard();
}

void loop() {
  if (sim.available() > 0){
     char c = sim.read();
      if (isPrintable(c)){
      Mess.concat(c);
      }else if (c == '\n'){
```

```
        Parameter = Mess;
      Mess = "";
      Serial.println(Parameter);
      var1 = Parameter.indexOf("Stop");


    }
      if (var1 == 0){
        digitalWrite(buzzer, LOW);
        digitalWrite(RELAY_PIN, LOW);
        var1=1;
       }
  }
  network.update();
  while ( network.available() ) {     // Is there any incoming data?
    RF24NetworkHeader header;
    unsigned long incomingData;
    network.read(header, &incomingData, sizeof(incomingData)); // Read the
incoming data
    Serial.println(incomingData);  // tell servo to go to a particular angle
    if(incomingData == 100){
      y = 0;
      if (x == 0){
        x = 1;
        SendMessage(Contact_01);
      }
      digitalWrite(buzzer, HIGH);
```

```
    digitalWrite(RELAY_PIN, HIGH);
  }

 }
  y = y + 1;
  if (y == 10){
    x = 0;
    digitalWrite(buzzer, LOW);
    digitalWrite(RELAY_PIN, LOW);
  }

  delay(1000);

}
//*******************************SD card Start*********************************
void getSDcard(){
  // Reading the file Contact
 myFile = SD.open("contact1.txt");
 if (myFile) {
  Serial.println("Contact:");
  // Reading the whole file
  while (myFile.available()) {
    //Serial.write(myFile.read());
    char c = myFile.read();
    if (isPrintable(c)){
```

```
    Bucket.concat(c);
    }else if(c == '\n'){
      data = Bucket;
      Bucket = "";
      Serial.println(data);
      Contact_01 = data;
    }
  }
  myFile.close();
}
else {
  Serial.println("error opening test.txt");
}

// Reading the file of Location
myFile = SD.open("location.txt");
if (myFile) {
  Serial.println("Location:");
  // Reading the whole file
  while (myFile.available()) {
    //Serial.write(myFile.read());
    char c = myFile.read();
    if (isPrintable(c)){
      Bucket.concat(c);
    }else if(c == '\n'){
```

```
    data = Bucket;
      Bucket = "";
      Serial.println(data);
      Location = data;
    }
  }
  myFile.close();
 }
 else {
  Serial.println("error opening test.txt");
 }

 }
//*******************************SD card End*********************************

//***********************************SMS Start*******************************
void SendMessage(String number)
{
 Serial.println ("Sending SMS");
 sim.println("AT+CMGF=1");    //Sets the GSM Module in Text Mode
 delay(200);
 //Serial.println ("Set SMS Number");
 sim.println("AT+CMGS=\"" + number + "\"\r"); //Mobile phone number to
send message
 delay(200);
```

```
  String SMS = "Rugdoll is activated at the location
www.google.com/maps/place/" + String(Location) + "\n you can check to
assist the situation or let the machine do her job.";
  sim.println(SMS);
  delay(100);
  sim.println((char)26);// ASCII code of CTRL+Z
  delay(200);
  _buffer = _readSerial();
}
void RecieveMessage()
{
  Serial.println ("SIM800L Read an SMS");
  sim.println("AT+CMGF=1");
  delay (200);
  sim.println("AT+CNMI=1,2,0,0,0"); // AT Command to receive a live SMS
  delay(200);
  Serial.write ("Unread Message done");
}
String _readSerial() {
  _timeout = 0;
  while  (!sim.available() && _timeout < 12000  )
  {
    delay(13);
    _timeout++;
  }
  if (sim.available()) {
```

```
    return sim.readString();
  }
}
void callNumber(String number) {
  sim.print (F("ATD"));
  sim.print (number);
  sim.print (F(";\r\n"));
  _buffer = _readSerial();
  Serial.println(_buffer);
}

//**********************************SMS End**********************************
```

APPENDIX H

CURRICULUM VITAE

JOMAR F. GODINEZ

Address: Brgy. San Jose, Tandag City, Surigao del Sur
Contact #: 09551688249
EmailAddress: jomargodinez70@gmail.com

OBJECTIVES

Find a job in the field of computer science that aligns with my interests and career goals. I want to build a strong foundation in computer science by reviewing and strengthening my knowledge of the core concepts and programming languages.

PERSONAL INFORMATION

Age: 27
Date Of Birth: November 3, 1995
Bplace: Cebu City
Institutional Affiliation: BSCS
Fathers Name: Marciano Godinez
Mothers Name: Rosalina Godinez

Gender: Male
Civil Status: Single
Citizenship: Filipino
Religion: Roman Catholic
Occupation: None
Occupation: None

EDUCATIONAL BACKGROUND

College: North Eastern Mindanao State University S.Y. 2022-2023
Course: Bachelor of Science in Computer Science

Secondary: Lugait National High School S.Y. 2019-2020
Elementary: Mactan Elementary School S.Y. 2011-2012

SKILLS

- Programmer
- Microsoft, Power Point and Excel Literate
- Photo and Video Editing
- Graphic Designer

JEICER JHON D. GONJORAN

Address: San Miguel, Surigao del Sur
Contact #: 09518155237
Email Address: gonjoranjj18@gmail.com

OBJECTIVES

In search of a demanding job in a reputable company where I can develop new skills, increase my knowledge, and apply what I've learned. to be given a chance to use my abilities to their fullest and advance the organization.

PERSONAL INFORMATION

Age: 23
Date Of Birth: March 23, 2000
Bplace: Brgy. Sagbayan, San Miguel, SDS
Institutional Affiliation: BSCS

Gender: Male
Civil Status: Single
Citizenship: Filipino
Religion: Jehovah's Witness

Fathers Name: Joel A. Gonjoran
Occupation: Farmer
Mothers Name: Marcelina D. Gonjoran
Occupation: Housekeeper

EDUCATIONAL BACKGROUND

College: North Eastern Mindanao State University S.Y. 2022-2023
Course: Bachelor of Science in Computer Science

Secondary: Sagbayan National High School S.Y. 2018-2019
Elementary: Pedro Meranda Elementary School S.Y. 2012-2013

SKILLS

- Computer Skills
- Communication Skills
- Graphic Designing
- Photo Editing using Photoshop

LEA MAE V. LINAZA

Address:
Contact #: 09076792039
Email Address: leamaelinaza@gmail.com

OBJECTIVES

To get an opportunity where I can make the best of my potential and contribute to the organization's growth.

PERSONAL INFORMATION

Age: 24
Date Of Birth: January 3, 1999
Birth Place: Lingig, Surigao Del Sur
Institutional Affiliation: BSCS
Fathers Name: Ysmael A. Linaza
Mothers Name: Leecil A. Linaza

Gender: Female
Civil Status: Single
Citizenship: Filipino
Religion: Catholic
Occupation: Fisherman
Occupation: Housewife

EDUCATIONAL BACKGROUND

College: North Eastern Mindanao State University S.Y. 2022-2023
Course: Bachelor of Science in Computer Science

Secondary: Lingig National High School S.Y. 2019-2020
Elementary: Tomas Sim Elementary School S.Y. 2011-2012

SKILLS

♦ Communication Skills

♦ Computer Literate

JENJEN M. MACA

Address: Carromata, San Miguel, Surigao del Sur
Contact #: 09462435745
Email Address: jenjenmaca30@gmailcom

OBJECTIVES

To obtain a position that will effectively utilize my skill and the opportunity to work with a team in return, I offer my service and determination to be an asset to your company thought the duration of my training period.

PERSONAL INFORMATION

Age: 23
Date Of Birth: March 30, 2000
Bplace: Bagyang, San Miguel, Sds
Institutional Affiliatio: BSCS
Fathers Name: Garcia M. Maca
Mothers Name: Norma M. Maca

Gender: Female
Civil Status: Single
Citizenship: Filipino
Religion: Inc
Occupation: Farmer
Occupation: Housewife

EDUCATIONAL BACKGROUND

College: North Eastern Mindanao State University S.Y. 2022-2023
Course: Bachelor of Science in Computer Science

Secondary: San Miguel National High School S.Y. 2018-2019
Elementary: Bagyang Elementary School S.Y. 2011-2012

SKILLS

♦ Basic Web Design
♦ Team and Collaboration

JEROME MACHECA

Address: Marihatag, Surigao del Sur
Contact #: 09279775250
Email Address: frikk48@gmailcom

OBJECTIVES

Chase your dream and work hard.

Personal Information

Age: 24
Date Of Birth: June 09, 1998
Bplace: Marihatag, Surigao Del Sur
Institutional Affiliation: BSCS
Father's Name: Reynaldo U. Macheca
Mother's Name:Ma. Dioneda P. Macheca

Gender: Male
Civil Status: Single
Citizenship: Filipino
Religion: Baptist
Occupation: Farmer
Occupation: House Wife

EDUCATIONAL BACKGROUND

College: North Eastern Mindanao State University S.Y. 2022-2023
Course: Bachelor of Science in Computer Science

Secondary: Holy Child Academy S.Y. 2018-2019
Elementary: Marihatag Central Elementary School S.Y. 2011-2012

SKILLS

♦ Proficient in Microsoft Word, Power point
♦ Photo Editing using Photoshop

RIALYN N. MAGALLANES

Address: Prk. Golden Hills Tandag City, SDS.
Contact #: 09709156655
Email Address: rialyn.magallanes@gmailcom

OBJECTIVES

Fresh college graduate looking to obtain a position as a front desk receptionist. I'd like to utilize my high degree of work efficiency and organizational skills contribute to the overall success of the organization.

PERSONAL INFORMATION

Age: 23
Date Of Birth: November 29, 1999
Bplace: Binangonan Rizal, Antipolo City
Institutional Affiliatio: BSCS
Fathers Name: Regino B. Magallanes
Mothers Name: Dailenda R. Nueva

Gender: Female
Civil Status: Single
Citizenship: Filipino
Religion: Assembly of God
Occupation: Deceased
Occupation: House Keeper

EDUCATIONAL BACKGROUND

College: North Eastern Mindanao State University S.Y. 2022-2023
Course: Bachelor of Science in Computer Science

Secondary: Buenavista National High School S.Y. 2018-2019
Elementary: Pangi Elementary School S.Y. 2011-2012

SKILLS

- ♦ Communication Skills
- ♦ Leadership
- ♦ Computer Literate
- ♦ MS Oriented